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International Conference on
AI-Driven Solutions for
Sustainable Smart Cities:
Challenges and Opportunities

09-10 April 2026



ADSSSC

2026



International Conference on AI-Driven Solutions for Sustainable Smart Cities: Challenges and Opportunities

9–10 April 2026, <https://adsssc.sahrdaya.ac.in>

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Indian Institute of Space Science and Technology (IIST), Trivandrum

To: the organizing committee of ADSSSCE-2026

Subject: Support letter for ADSSSCE-2026 conference

It is my great pleasure to extend warm greetings to all participants, authors, reviewers, and organizers of this prestigious conference, ADSSSCE 2026. On behalf of the IEEE Kerala Section, I commend the organizing committee for bringing together a vibrant platform that fosters innovation, collaboration, and the dissemination of cutting-edge research.

Sahrdaya is a leading autonomous institution affiliated with APJ Abdul Kalam Technological University, approved by AICTE, and accredited by NBA, NAAC, and IET. The college is also recognized as a Scientific and Industrial Research Organization (SIRO) by the Government of India. With a strong focus on academic excellence, research, and innovation, Sahrdaya has established several Centers of Excellence in collaboration with industry leaders such as GE, Accenture, and Agaphe.

This conference reflects the spirit of IEEE in advancing technology for humanity. The contributions presented in these proceedings demonstrate the commitment of our research community toward addressing real-world challenges through scientific and engineering excellence. It is particularly encouraging to see interdisciplinary efforts and emerging themes that align with global technological transformations, including AI-driven systems, next-generation communication networks, sustainable technologies, and intelligent infrastructure spanning Electrical, Electronics, Computer Science, Biomedical, Biotechnology, and Information Sciences.

IEEE Kerala Section has consistently strived to nurture a strong ecosystem of students, researchers, and professionals. Conferences such as this play a pivotal role in strengthening this ecosystem by enabling knowledge exchange, professional networking, and collaborative research opportunities.

I congratulate all authors whose work has been included in these proceedings and appreciate the dedicated efforts of reviewers and organizers in maintaining the highest standards of quality. I am confident that the insights and innovations shared here will inspire further research and contribute significantly to technological advancement.

I wish all the best for the organizing team, participants, and the college that took the initiative for the conference.

I wish the conference continued success and all participants a highly engaging and enriching experience

Kindest Regards,

Prof. B. S. Manoj

Chair, IEEE Kerala Section

Senior Professor & Associate Dean

Indian Institute of Space Science and Technology (IIST), Trivandrum



From the Desk of Executive Director

It gives me immense pleasure to convey my good wishes for the publication of the proceedings of the International Conference on AI-Driven Solutions for Sustainable Smart Cities: Challenges and Opportunities. This compendium represents significant research contributions addressing key challenges in urban environments through innovative integration of artificial intelligence and sustainable development. I am confident that it will serve as a valuable resource for researchers, practitioners, and policymakers working towards smarter and more sustainable cities.

I extend my sincere appreciation to the authors for their significant contributions and to the reviewers for ensuring the quality of this publication through their scrupulous evaluation. I also express my gratitude to the organizing committee for their dedicated efforts in bringing out these proceedings successfully.

I hope this compilation inspires further innovation and collaboration in the field.

Fr. Dr. Anto Chungath
Executive Director
Sahridaya College of
Engineering & Technology
Kodakara



From the Desk of the Principal

It gives me immense pleasure to welcome you to the International Conference on AI-Driven Solutions for Sustainable Smart Cities: Challenges and Opportunities (ADSSSC 2026), hosted by Sahridaya College of Engineering and Technology (Autonomous) in technical association with the IEEE Kerala Section.

As we stand at the threshold of a new era defined by Industry 4.0, the integration of physical and digital worlds is no longer a futuristic concept but a present-day necessity. This conference serves as a vital platform for researchers and practitioners to explore intelligent computational systems that foster inclusive, resilient, and sustainable urban environments across critical domains such as healthcare, mobility, and energy.

At Sahridaya, we believe in a vision of education through service.". This commitment is best exemplified by our GramVista initiative—a village-scale Cyber-Physical Systems (CPS) living laboratory aligned with the Unnat Bharat Abhiyan. By applying technologies like IoT, AI, and Digital Twins to solve real-world challenges in rural Kerala, we are ensuring that our technological advancements have a direct, positive impact on society.

This conference is also a celebration of our institutional growth. As an autonomous institution, we are pioneering new academic frontiers with the introduction of specialized programs in Cyber-Physical Systems (CPS) and AI/ML. We aim to produce Master System Integrators"—innovators who can navigate the complexities of both hardware and software to build the autonomous systems of tomorrow.

The diverse range of tracks in ADSSSC 2026, from AI-enabled Energy Systems to Sensors and Edge Intelligence, reflects the interdisciplinary mastery required to address

global challenges. We are honored to host distinguished keynote speakers and authors from across the globe to share their cutting-edge research.

I extend my sincere gratitude to the Organizing Committee, the Technical Program Committee, and our partners at IEEE for their tireless efforts in bringing this event to fruition. I am confident that the deliberations over these two days will spark new collaborations and inspire innovative solutions for the sustainable smart cities of the future.

I wish all the delegates a productive and enriching experience at Sahrdaya.

Dr. Ramkumar S.
Principal
Sahrdaya College of
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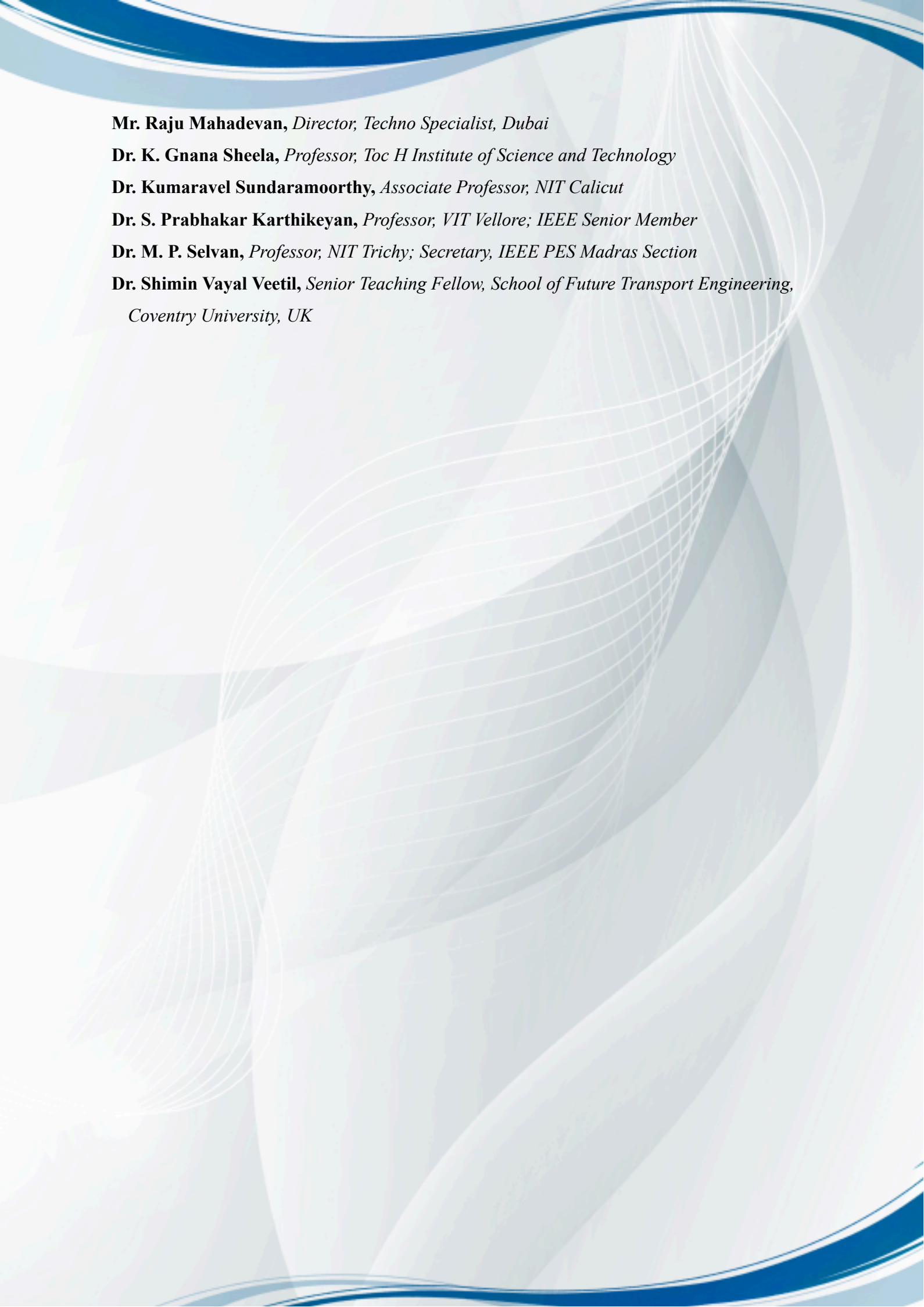
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About Sahridaya

Sahridaya College of Engineering & Technology (Autonomous) is a leading engineering institution in Kerala with a strong focus on academic excellence, research, innovation, and industry collaboration. The college offers flagship B.Tech programs in Biotechnology, Biomedical, Civil, Computer Science, Electrical & Electronics, and Electronics & Communication, along with M.Tech programs in Embedded Systems, Industrial Biotechnology, and Computer Science. Sahridaya is accredited by NBA and NAAC, recognized as a Scientific and Industrial Research Organization (SIRO) by DSIR, and hosts several centers of excellence and innovation hubs that support startups, entrepreneurship, and translational research.

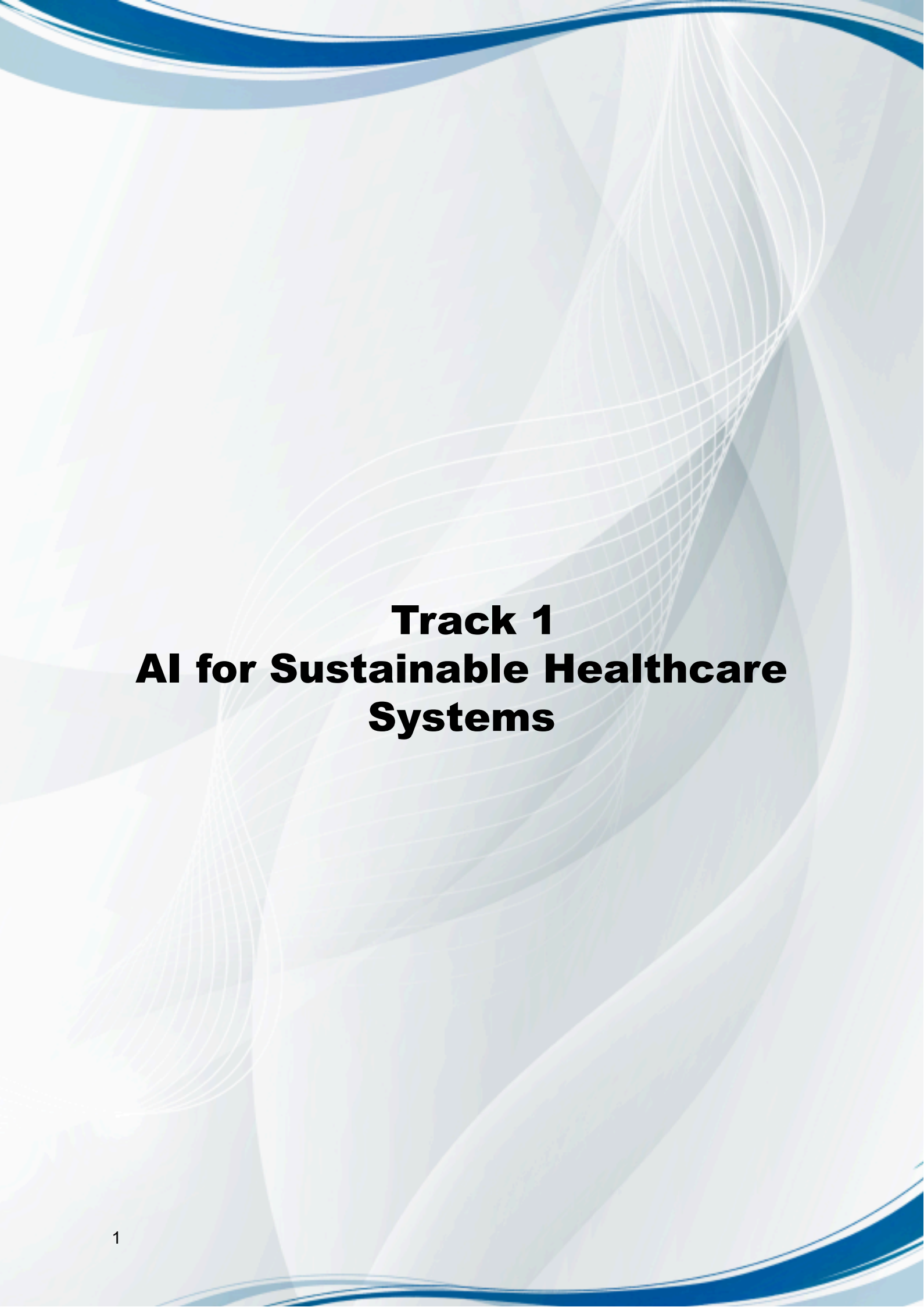
About the Conference

The International Conference on AI-Driven Solutions for Sustainable Smart Cities: Challenges and Opportunities (ADSSSC 2026) focuses on intelligent computational systems that enable inclusive, resilient, and sustainable urban environments across domains such as healthcare, mobility, energy, environment, and infrastructure. The conference brings together researchers, practitioners, and industry professionals to present cutting-edge research, innovative applications, and cross-disciplinary solutions in Electrical, Electronics, Computer Science, Biomedical, Biotechnology, and Information Sciences.

ADSSSC 2026 features invited keynotes, technical paper sessions, tutorials, and workshops, and aims to foster strong collaborations between academia, industry, and government agencies.

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Track 1

AI for Sustainable Healthcare Systems

AI-Powered Brain MRI Assistant for Automated Tumor Segmentation, Report Generation, and Treatment Prediction

Mohamed Saneen, Adithya Anil, Aarsh Balakrishnan, Nyju S, Mahesh K M

Computer Science and Business Systems

Rajagiri School of Engineering and Technology, Kochi

Abstract: Radiologists interpret numerous brain MRI scans on a regular basis and generate lengthy diagnostic reports that are time-consuming and liable to human error, leading to delayed therapy and additional healthcare cost. To resolve these issues, we introduce an AI-powered Brain MRI Analysis System that will automate tumor segmentation, reporting, and the prediction of therapy, as a decision-support tool and not a replacement. With the use of deep learning models with precise tumor delineation and predictive analysis, coupled with natural language generation for report automation, the system enables clinicians to review and validate results via an interactive system. The process decreases workload, enhances diagnostic accuracy, and personalizes therapy suggestions, thus optimizing clinical efficiency and patient outcomes. The system also aids less experienced clinicians by enhancing their diagnostic confidence and decision-making skills.

From Algorithms to Heartbeats: A Survey of AI and IoT Approaches in Heart Disease Prediction

Shruthi S Nair, Sreelakshmi S, Sathya U, Kiruthika S S,
Saveetha Engineering College, Chennai

Abstract: Cardiovascular disease is one of the leading causes of morbidity and mortality globally, without regard to geography or social status. Advancements in Internet of Things (IoT) as well as artificial intelligence (AI) have opened up new avenues of proactive cardiovascular care especially in the space of early detection continuous monitoring as well as risk assessment. In this review, we collect and analyze current AI and IoT based heart disease prediction frameworks, with an emphasis onto the endeavors to translate from the lab to the clinic. The state of the art is reviewed in terms of approaches from traditional machine learning, deep learning architectures, and wearable sensor-based systems as well as their potential in decision support and patient management. From a technical and clinical perspective, practical issues of data security and privacy, algorithms transparency and integration with the existing healthcare workflows are explored. To aid in developing the field, datasets, evaluation metrics and typologies of methods are summarized as a reference. The review also highlights research gaps and new directions of work that might contribute to more trustworthy, personalized and ethically sensitive intelligent systems in cardiovascular health care.

A Lightweight Residual Quantum Deep Learning Model for Grading of Age-related Macular Degeneration

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²Department of ECE, National Institute of Technology Tiruchirappalli

Abstract: Age-related macular degeneration is a leading cause of persistent central vision loss in elderly populations worldwide. AMD affects nearly 9% of the world population. Hence, an early and accurate detection is vital to slowing disease progression. But the manual analysis is time-consuming, complex, and subjective. This paper introduces a new lightweight quantum residual deep learning model that classifies optical coherence tomographic images into dry AMD, wet AMD, or healthy. The proposed hybrid architecture uses standard convolutional layers for initial feature extraction, which is followed by innovative residual quantum-inspired blocks. These blocks use quantum principles to improve feature extraction. These principles are implemented through tanh activations and unbiased linear transformations. This equips the network to extract complex, non-linear correlations within the images. The model achieved 98.53% accuracy and an AUC value of 0.9998 on the Mendeley dataset. Also, another dataset (NEH) is used to test the model's ability. The test results show the model's ability to produce consistent results even on datasets not used for testing. The architecture displayed a low computational cost by employing Global Average Pooling (GAP) with a lower parameter count. With reduced computational cost, the proposed model can be a suitable implementation on edge devices for realtime diagnosis.

NUTRIVISION: A System for Personalized Nutrition using Machine Learning to Identify Vitamin Deficiency

Unnikrishnan S. Kumar, Ali Askar K. M, Nisna Haneefa K, Abhiram T, Saniya S, Anagha C
Department of CSE, Jawaharlal College of Engineering and Technology

Abstract: Vitamin deficiency remains a global public health concern, leading to severe health complications if not detected early. Early detection is a prerequisite for successful treatment and upcoming image processing and neural network algorithms can give accurate diagnosis. This system focuses on the use of image processing and neural network approaches to detect signs of deficiencies in vitamins from visual non-invasive data, such as nails, eyes, tongue, and skin. We offer a complete theoretical investigation of the prevailing strategies and address the benefits along with limitations arising through neural network architectures, e.g., Convolutional Neural Networks (CNN), ResNet, DenseNet in feature shaping and categorization. The issues surrounding common complaints such as data scarcity, variability and model accuracy are examined here as well. The proposed system demonstrates the potential of deep learning for non-invasive vitamin deficiency detection.

Adaptive Residual Network Optimization for Accurate Diabetic Foot Ulcer Image

Classification

G. M. Sathyaseelan ¹, M. Rajalakshmi ², Nandhitha S ³, Preethi M ⁴, S. P. Santhoshkumar ⁵

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³*Vivekanandha College of Engineering for Women*

⁴*Vel Tech High Tech Dr.Rangarajan Dr.Sakunthala Engineering College*

⁵*Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology*

Abstract: Diabetic foot ulcer (DFU) is one of the severe complications of diabetes mellitus which can lead to severe infection and limb amputation if not detected at an early stage and conventional clinical evaluation is also time-consuming and is likely to exhibit inter-observer effects. To deal with such issues, this paper proposes an Adaptive Residual Network Optimization (ARNO) framework for automated diabetic foot ulcer image classification of DFU that is specifically aimed at addressing the issues of low and unequal data samples. The suggested solution combines the deep residual learning models with the adaptive optimization techniques to enhance the feature discrimination and reduce overfitting and inter-class bias. Several state of the art convolutional backbones, such as ResNet variants and lightweight transfer learning networks are explored to find the best feature representations, which can be supplemented by image enhancement methods as well as data augmentation to address the difference in illumination, skin texture and ulcer appearance. This framework raises both binary and multi-class DFU severity classification, which succeeds in distinguishing healthy skin and the various grades of ulcers and experimental analyses show better behavior in accuracy, precision, recall, and F1-score than traditional deep learning methods, and maintains computational efficiency, and hence the appropriateness of the framework in scalable clinical decision-support systems and operation in resource-limited healthcare settings.

Generative Artificial Intelligence Vs. Transformer-Based Framework for Classification of Cardiovascular Disease Patients using Transcriptomic Gene Data

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Abstract: The classification of cardiovascular diseases using gene expression data is challenging due to high dimensionality, noise, and limited sample sizes. Traditional machine learning and deep learning models often struggle to generalize effectively in such settings, motivating the need for generative and energy-based approaches. We propose a Bio-GenAI framework that leverages generative artificial intelligence with denoising score matching to learn the underlying distribution of gene expression data. Noise is added to real samples, and the model learns to progressively restore them toward the real data manifold. An energy-based scoring mechanism is then used to classify case and control subjects. The proposed Bio-GenAI framework demonstrates improved classification performance compared with transformer-based models. The framework is evaluated on Hypertrophic Cardiomyopathy (HCM) and Acute Myocardial Infarction (AMI) datasets using K-fold validation. The proposed Bio-GenAI model achieved ~98% accuracy with an AUC of 0.99 in seen paradigm. In Unseen paradigm, ~94% accuracy with an AUC of 0.99, outperforming Transformers by 3.14% in HCM data and 5.2% in AMI data. All the models meet the regulatory requirements having <10% difference between seen and unseen paradigm. The Bio-GenAI framework demonstrates robust performance and generalization, highlighting its potential for reliable gene-based cardiovascular disease classification.

Efficient Hybrid Features with Ensemble Normalization for Cross-Subject EEG Emotion Recognition

Praveena S Kammath¹, Dr. G R Gnana King²

¹SCMS School of Engineering and Technology,

²Sahrdaya College of Engineering and Technology

Abstract: EEG based emotion recognition exhibits poor cross-subject generalization with state-of-the-art accuracies typically ranging 65–72% on benchmark datasets. This paper proposes an efficient hybrid feature set combining band-power, Hjorth parameters, differential entropy (DE), and alpha-band asymmetry. These features are refined using a novel ensemble normalization strategy to reduce inter-subject variability while preserving emotion-discriminative structure. Applied to 32 subjects each from the FACED and DEAP datasets, the pipeline employs 2-second sliding windows after 1–45 Hz bandpass filtering. XGBoost classification achieves 77.2% accuracy on FACED and 69.2% on DEAP using window-wise cross-validation, demonstrating robust cross-dataset generalization. To further address evaluation rigor and potential data leakage, we additionally report results under a stricter random split protocol, yielding approximately 48–50% accuracy, which reflects realistic cross-subject generalization without window overlap effects. Ablation studies confirm that the hybrid feature combination outperforms individual components, with full ensemble normalization critical to performance. This work significantly improves EEG-based emotion recognition by providing balance between computational efficiency and accuracy across different users, offering a scalable baseline for developing real-world affective computing applications.

An Interpretable Graph Attention Network for Sleep Stage Classification Using Statistically Validated Phase Amplitude Coupling

Vrinda V. Gopal T¹, Dr. G R Gnana King²

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²*Department of ECE, Sahridaya College of Engineering and Technology*

Abstract: Automatic sleep stage classification using single-channel EEG remains challenging due to nonlinear temporal dynamics and complex cross-frequency interactions. This paper proposes a physiologically interpretable framework that integrates statistically validated Phase–Amplitude Coupling (PAC) with Graph Attention Networks (GAT) for three-class sleep staging (Wake, N3, REM). PAC is computed using Tort’s Modulation Index with surrogate-based statistical validation to retain only significant cross-frequency interactions. The resulting PAC matrix is modeled as a bipartite graph and processed using a multi-head GAT architecture. Additionally, nonlinear temporal dynamics are captured using Visibility Graph (VG) features. The hybrid spectral–temporal representation improves discriminative learning while maintaining interpretability. Evaluation on the Sleep-EDF dataset demonstrates that the proposed framework achieves 76.19% classification accuracy with macro F1 of 0.66, outperforming conventional PAC-based multilayer perceptron and CNN baselines using identical single-channel input. Attention analysis reveals physiologically consistent stage-specific coupling patterns, highlighting the interpretability and effectiveness of the proposed approach.

ALERA: An AI-Powered Conversational Skin Disease Classification and Allergen Detection System Using Deep Learning and Retrieval-Augmented Generation

Midhun G Kaimal, Kailas MS, Mohammed Roshan A N, Joshna Jojo, Anugraha K R
Dept. of Computer Science and Eng, Sahridaya College of Eng. and Tech.

Abstract: ALERA (Allergy Skin Health Intelligence) is an AI-assisted clinical decision-support system for skin disease and allergen-related risk assessment. The framework integrates lesion-image classification using ResNet50 with a Retrieval-Augmented Generation (RAG) chatbot. The conversational module uses a Mistral large language model served through the Groq API to generate context-aware, patient-friendly guidance grounded in trusted medical sources. The system combines image analysis and conversational support to provide interpretable diagnostic assistance in telemedicine workflows. Experimental observations indicate strong classification performance and low-latency responses, supporting timely and accessible preliminary dermatological guidance.

CTG Based Fetal Health Classification Using ML

Anjana Varghese, Anlin Davis, Anly C T, Annmariya P S, Anil Antony

Sahrdaya College of Engineering and Technology

Abstract: With Interpretability Using SHAP Manual interpretation of cardiotocography (CTG) is often subjective, leading to inconsistent fetal health assessments. To address this, we propose a stacked generalization ensemble using a Bayesian Neural Network (BNN) as the meta-learner and SVM, Random Forest, XGBoost, and CatBoost as base learners. Class imbalance is handled using SMOTE within a crossvalidation framework to generate robust meta-features, which are combined with scaled CTG inputs. Model transparency is ensured through real-time SHAP-based explanations. Evaluated on the UCI CTG dataset, the proposed ensemble achieves a state-of-the-art accuracy of 95.30% and a balanced accuracy of 93.30%, consistently outperforming all. Additionally, an exploratory multimodal framework is introduced, incorporating CNN-based CTG image analysis and hybrid fusion of visual and numerical features, highlighting a promising direction toward end-to-end fetal health diagnostics.

Improving Doctor-AI Collaboration Using Heatmap Based Explainable Medical Image

Models

M.Premchander, Kunja Poornitha, Kurukuntla Shreya, Mothe Shalini, Kola Anusha,
Kolipaka Bhavya Sri,

Department of IT Mallareddy Engineering College for Women

Abstract: Machines now catch illness in pictures of insides - X-rays, MRI views. Many clever programs keep their logic locked away. What shows up on screen comes with no explanation. That missing piece? It chips away at trust. When one teammate stays silent, working together gets harder. Out of nowhere, heatmaps show what the machine notices. Bright patches point to spots that matter, pulling attention like gravity. When logic steps into the light, belief follows. Where uncertainty lingered, clarity now connects. Noticing shapes meaning as much as results do. Not left to assumptions, the system highlights areas it paid extra attention to. These markings appear thanks to a technique exposing key influences behind each choice. Medical staff observe precisely which regions the program studied closely. Clarity comes through visibility into the steps leading up to judgment. Understanding builds when people trace the path the machine took. Noticeable is the way this approach finds problems in scans and marks their locations precisely. Not merely delivering an outcome, it underlines regions a physician would prioritize. Open-access chest X-rays were used for testing - split between healthy and abnormal categories. A large chunk trained the system, a portion adjusted its settings, another batch verified results. A solid performance showed up in the numbers. Accuracy landed at 94.2 percent. Wrong alerts happened rarely. Nearly every real instance got caught. Results came through clearly. What stands out is how heatmaps reveal the thinking behind choices. Seeing where the system pays attention builds confidence over time. Clinicians lead the process, supported by clues rather than sidelined. When logic shows its face early, fewer odd outcomes pop up later.

Neonatal Care System For Jaundice And Congenital Heart Disease

Felix Bobbin, Jiyad T J, Joel Jose, Sania Biju, Jayalakshmi P K

Department of BME, SCET

Abstract: To offer efficient neonatal health screening in resource-poor settings, small, inexpensive, and smart monitoring systems are necessary. In this project, a real-time and integrated neonatal monitoring and care system that combines multimodal physiological sensing with environmental control is presented. The system captures single-lead ECG signals to identify the heart rate and heart rate variability, which is used to detect early signs of hemodynamic dysfunction associated with CHD. Until now, concurrently, photoplethysmography (PPG) data obtained with the MAX30102 sensor are utilized to estimate vascular stiffness based on pulse transit time (PTT) and pulse wave velocity (PWV) and oxygen saturation. Moreover, an RGB camera takes snapshots of the skin to estimate bilirubin concentration in order to screen jaundice in a non-invasive manner. All data streams are processed in Raspberry Pi and displayed on interactive touchscreen dashboard, and logs are kept to analyse them offline. The system communicates with relay-controlled incubator units to control phototherapy and temperature regulation which allows automatic recommendations and clinician alerts.

Real-Time Personalized Air Quality Health Risk Monitoring

Ameena K.S, Anna Lin Noby, Ann Mariya C.R, Binisha P.B, Jyothsna G.K

Department of Computer Science and Engineering, Jyothi Engineering College

Abstract: Air quality is hazardous and poses significant health risks, particularly for individuals with respiratory and cardiovascular conditions. Existing systems provide generalized information based on predefined thresholds and do not account for individual health status or physiological responses. This paper presents a real-time Internet of Things (IoT)-based framework for personalized health and environmental monitoring. The proposed system collects environmental parameters, including air quality, temperature, and humidity, along with physiological data such as heart rate and oxygen saturation. Machine learning techniques are applied to analyze the collected data and assess potential health risks associated with air pollution exposure. The system also recommends safer routes based on current air quality conditions, enabling informed decision-making. By integrating environmental and personal health data, the proposed approach enhances individual awareness and supports proactive health management. This framework contributes to improved personal health monitoring and effective environmental risk mitigation.

NeoPulse: Fetal Health Monitoring System

Aiswarya K, Aleena Pauly, Antony Peter, Emmanuel Jiju, Binet Rose Devassy,
Dr. G R Gnana King,

*Department of Electronics and Communication Engineering, Sahrdaya College of
Engineering and Technology*

Abstract: Continuous fetal health assessment is critical for early detection of pregnancy related complications. Existing monitoring systems are primarily hospital-based and do not support integrated real-time evaluation of both fetal heart rate and movement in home environments. This paper presents NeoPulse, a non-invasive fetal monitoring framework that combines multimodal signal acquisition, robust signal processing and supervised classification for automated fetal condition assessment. Acquired physiological signals are processed using digital filtering techniques to suppress noise and artifacts, followed by time- and frequency-domain feature extraction to characterize fetal cardiac and movement patterns. A classification model is implemented to identify normal and abnormal fetal conditions based on extracted features. Experimental evaluation demonstrates reliable signal quality enhancement and effective classification performance under controlled validation settings. The proposed framework supports real-time monitoring through a mobile interface and provides automated alert generation. The results indicate that the system offers a portable and efficient approach for continuous prenatal monitoring in both clinical and home-based applications.

Virtual Screening And Molecular Dynamics Analysis Of Natural Inhibitors For Liver Cirrhosis With Emerging Ai Applications

Joanna Joseph Tharakan , Joshna B Ann , Shivani Santhosh , Dr. Ranimol G ,
Department of Biotechnology, Sahrdaya College of Engineering & Technology

Abstract: Liver cirrhosis is a progressive and irreversible chronic liver disorder characterized by sustained hepatocellular injury, inflammatory signaling, oxidative stress, and extensive deposition of extracellular matrix components. Existing therapeutic strategies slow disease progression but fail to reverse fibrosis, highlighting the need for novel antifibrotic agents. Natural phytochemicals have gained scientific interest due to their strong hepatoprotective, antioxidant, and anti-inflammatory activities. In this study, five natural compounds—kaempferol, silybin, curcumin, luteolin, and caffeic acid—were evaluated through a comprehensive in-silico workflow comprising molecular docking, pharmacokinetic prediction, toxicity assessment, and molecular dynamics simulation. Kaempferol exhibited the most favorable pharmacokinetic parameters, the safest toxicity profile, and the strongest and most stable molecular interactions. Docking analysis further revealed that kaempferol displayed a distinctively high affinity toward a key fibrogenic protein, suggesting significant inhibitory potential. Molecular dynamics simulation confirmed the stability of the interaction under physiological conditions. Furthermore, the findings highlight the potential of integrating artificial intelligence-based drug–target interaction prediction models in future studies to enhance screening efficiency and accelerate antifibrotic drug discovery. Overall, kaempferol emerged as a promising natural molecule with potential antifibrotic activity, warranting further in vitro and in vivo investigation.

Comparative Analysis of Machine Learning Algorithms for Spine Condition

Classification: A Multi-Strategy Evaluation Framework

Sindhura D N , Radhika M Pai , Manohara Pai M M,

Manipal Institute of Technology

Abstract: In this study, an in-depth comparison between classical machine learning algorithms and Automated Machine Learning (AutoML) systems in spine condition classification based on a multi-strategy assessment model is done. The dataset on vertebral column from the UCI Machine Learning Repository includes 310 samples of patients with six biomechanical variables is used. Eight classical algorithms including Logistic Regression, Support Vector Machine, Random Forest, Gradient Boosting, KNearest Neighbors, Naive Bayes, Decision Tree, and CatBoost were compared with TPOT AutoML on a variety of validation approaches, such as 2-fold, 5-fold and 10-fold cross-validation, holdout validation with different test sizes (20, 30, 40), statistical significance testing with Welch t-test. Findings show that the overall performance of the Logistic Regression had the highest mean cross-validation accuracy of 85.48% and standard deviation of 4.89% and the shortest training time, which is significantly better than AutoML strategies. Random Forest had the highest consistency of best performers with minimum standard deviation ($\pm 4.40\%$). Classical algorithms easily surpassed AutoML (TPOT) for all strategies of validation, with TPOT taking more time to train, and with lower accuracy ($83.87\% \pm 5.86\%$). The results presented in the study suggest a algorithm selection strategy in medical classification problems using small to medium datasets and prove that more carefully tuned traditional algorithms are more efficient, accurate, and generalizable than AutoML solutions. Logistic Regression had the best accuracy of 85.48% in 0.06 seconds in 5-fold cross-validation, more than AutoML (TPOT) with an accuracy of 83.87% with significantly higher training time of 187.67 seconds.

An Optimized Attention-BiLSTM Framework for Polycystic Ovary Syndrome Classification with LIME-Based Explainability

Jilsy A J, Ardra V A, Jarin T

Department of Computer Science Engineering, Jyothi Engineering College

Abstract: Polycystic Ovary Syndrome (PCOS) is a multifactorial endocrine disorder that affects women of reproductive age and leads to metabolic, hormonal, and reproductive complications. Proper identification of PCOS from heterogeneous clinical and biochemical data is hard because of the complex interaction among features. To address this, an optimized Attention based Bidirectional Long Short-Term Memory (OA-BiLSTM) model is proposed for better feature extraction and classification of PCOS. The OA-BiLSTM model captures temporal and contextual dependencies among features; further, it shows the essential attributes by the attention mechanism. In addition, a meta-heuristic Sand Piper Optimization (SPO) is presented to fine-tune the parameters of the model and enhancing classification performance. For ensuring model transparency and interpretability, the Local Interpretable Model-Agnostic Explanations (LIME) is integrated for visualizing and explaining feature contributions in the decision-making process.

Hierarchical Multiview Dense Ensemble Model for Parkinson's Disease Prediction on T2-Weighted MRI

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Abstract: Parkinson's disease (PD) detection using MRI imaging has become popular due to its potential for early diagnosis. This study utilises T2-weighted Magnetic Resonance Imaging (MRI) to discover diseased features associated with PD. A novel deep learning framework based on the DenseNet121 with ensemble fusion convolutional neural network is designed to classify PD from T2-weighted MRI data, comprising axial, coronal, and sagittal slices. The dataset consists of 1428 MRI images of real and augmented images from multiple views (axial, coronal and sagittal) of brain MRI and is split into training (80%) and testing (20%) sets, with class imbalance addressed through computed class weights. Different models were trained for each slice orientation, global average pooling, dropout regularisation (0.5 and 0.3), and transfer learning with a partially frozen DenseNet121 backbone. To overcome overfitting, early stopping and learning rate reduction are employed. Predictions from the three pretrained models are ensembled using logistic regression meta-learner averaging to improve classification performance. Evaluation metrics calculated are accuracy, precision, recall, F1 score and AUC score. This model produces an AUC of 96% with an accuracy of 89.6%, which indicates that the proposed model outperforms the existing model.

A Blockchain-Integrated Deep Learning Framework for Privacy-Preserving Chest X-Ray Disease Diagnosis: A Comprehensive Survey

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Abstract: The growing use of deep learning for chest X-ray (CXR) analysis has significantly improved the early detection of pulmonary diseases. However, the increasing reliance on large-scale medical datasets has brought renewed attention to issues of privacy, data ownership, and the secure handling of sensitive clinical information. Conventional centralized data storage and model-training approaches expose healthcare institutions to risks such as unauthorized access, data tampering, and limited transparency in the transfer of diagnostic information. To address these concerns, recent research has explored the integration of blockchain technology with deep learning pipelines to create privacy-preserving and trust-enhancing diagnostic frameworks. This paper presents a comprehensive survey of blockchain-enabled deep learning models developed for CXR disease diagnosis. The review covers architectural designs, model-training strategies, secure data-sharing mechanisms, and privacy-preserving techniques including federated learning, differential privacy, and encryption-based computation. We highlight how blockchain can support decentralized collaboration between hospitals, enable traceable and verifiable model updates, and strengthen the reliability of automated diagnostic systems. The survey also examines several open-source datasets, performance trends, system bottlenecks, and technical considerations that influence real-world deployment. The findings indicate that blockchain integration can enhance trust, accountability, and security without compromising diagnostic accuracy, although challenges remain in scalability, regulatory compliance, and computational overhead. The paper concludes with key research gaps and future directions for developing secure, interoperable, and clinically adoptable CXR diagnostic frameworks that combine the strengths of decentralized technologies and advanced deep learning models.

Metaheuristic-Enhanced Hybrid Deep Learning for Accurate Early Alzheimer's Diagnosis

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Abstract: There is a growing interest in using deep learning techniques to automate the detection and classification of Alzheimer's Disease due to the increasing availability of large-scale multimodal datasets from neuroimaging techniques. A systematic review on the use of deep learning neuroimaging data for the AD diagnosis to find out other deep learning techniques was conducted. On this matter, the proposed method contains an effective classification of AD using multi-task learning with discriminative convolutional high-order Boltzmann Machine(DCBM) model with hybrid feature maps. DCBM was designed to go beyond the traditional deep Boltzmann machines (DBMs), which limited their use to feature extraction. The proposed method utilizes a multitask learning framework with identification and verification tasks to combat overfitting, which is more pronounced in the early phases of Alzheimer's due to high inter-subject variability and low intra-subject variability. The proposed DCBM system has two phases: first, learning from raw data through two-layer unstructured neural nets, applying dispersed filtering is performed; second, classifying the known elements with the SoftMax method. The framework surpasses known ensemble methods concerning accuracy with a clinical dataset from the National Alzheimer's Coordination Centre. The ensemble method with metaheuristic optimization showcases a remarkable accuracy achievement of 99.213%, underscoring the power of solving optimization problems when coupled with deep learning classification algorithms. For the future directions of deep learning for Alzheimer's disease (AD), it can focus on combining multi-omic layers of data and applying explainable AI approaches to shed light on critical Alzheimer's features and mechanisms to make AI model reasoning more transparent.

Secure AI-Driven Remote Medical Diagnostics for Skin and Retinal Imaging

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Abstract: Due to insufficient access to dermatology and ophthalmology specialists in remote and underserved communities, there is a need for trustworthy AI-assisted tele-diagnostic systems. This paper outlines the Mediscan framework—a secure remote diagnosis platform designed to conduct automated analysis and diagnosis from both skin lesion images and retinal fundus images. The proposed system includes a dual-stream deep learning architecture that utilizes a convolutional neural network (CNN) for skin lesion classification and a transformer-based Vision Transformer model for detecting retinal diseases. To enhance client transparency and clinician trust, the framework implements Class Activation Mapping (CAM) to provide visual feedback of AI-assisted model predictions. A required verification process via the human-in-the-loop approach mandates that each AI-assisted diagnosis undergo review by an appropriately-certified clinician prior to reporting, thereby ensuring that clinicians remain ultimately accountable for the diagnostic result. The proposed tele-diagnostic application utilizes a 3-tier architecture consisting of patient upload interface, a secure cloud-based inference engine, and a secure specialist validation dashboard as its primary components. Patient data security is achieved via protected communications protocols and encrypted storage. Each component has been tested using curated medical imaging datasets, and the proposed system has demonstrated reliable classification performance when evaluated against standardized metrics (accuracy, precision, recall, and AUC) when comparing to baseline models and shown no statistically significant differences for predictive consistency in the comparative analysis. The results validate the integration of deep learning, interpretability, secure data processing, and expert verification into one comprehensive telemedicine-based platform for real-world use within rural and underserved communities.

A Hybrid Multimodel Emotion Recognition Framework Integrating EEG and Facial Features with ANOVA-Based Feature Selection and Explainable AI

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Abstract: Multimodal signals for emotion detection enable a thorough examination of human affective states. In our work, we propose a hybrid-based approach that integrates emotion-specific electrical brain activity (electroencephalogram, EEG) and facial features to facilitate robust emotion recognition. EEG signals are analyzed using a form of neural network; Convolutional Neural Networks (CNNs) and a Transformer network are combined to create a modeling architecture that captures both spatial and temporal dependencies in brain signals. EEG signals contain rich temporal, spectral, and spatial information reflective of underlying neural activation; however, non-stationarity, noise artifacts, and subtle emotional patterns introduce significant challenges for feature extraction and classification. In this work, The proposed model achieved Accuracy of 0.98 using a Random Forest (RF) classifier integrated with ANOVA-based statistical feature selection is employed to identify highly discriminative EEG features while reducing dimensional complexity. For model transparency, two key XAI methods are employed: SHapley Additive exPlanations (SHAP) provides global feature importance by quantifying the average marginal contribution of each feature across the test set. Local Interpretable Model-agnostic Explanations (LIME) is used for local interpretability, generating simple, faithful explanations for individual predictions.

AMR-X: Design and Evaluation of an Integrated Machine Learning–Driven Antimicrobial Resistance Decision Support System

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Abstract: Antibiotic prescriptions often occur before laboratory results are available. Clinicians frequently rely on empiric therapy without access to local resistance data. We developed AMR-X, a system for estimating antimicrobial resistance risk from historical susceptibility records. The system requires only organism and antibiotic identifiers. Resistance was modeled as a binary classification task using XGBoost. The tool is deployed through a web interface and backend API. No genomic, demographic, or clinical variables were used in the model. On a test set of 281,364 records, ROC-AUC was 0.81 and sensitivity was 82%. Predictions include an evidence-strength indicator based on historical sample counts. AMR-X supports laboratory antimicrobial susceptibility testing rather than replacing it.

Intelligent Healthcare Platform For Doctor Appointments

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Abstract: The aim of this research is to create an intelligent doctor appointment system using machine learning algorithms to mitigate issues such as excessive wait, clinical misalignment and poor appointment management. This study defines a process of the development of a model based on four stages: (1) data collection and preprocessing such as the handling of missing values, (2) construction of the model based on the algorithms (Neural Network, Decision Tree, Logistic Regression and SVM), and (3) the assessment of the appointment datasets with the help of the constructed algorithms. Input variables for conducting doctor appointment scheduling will include doctor availability, specialty, consultation time, urgency of patient, and limitations imposed by the hospital. In contrast to existing solutions that fail to consider the mobility of variables such as cancellations or changes in demand, the intent of this model is to reduce overall wait times experienced by patients, improve utilization of resources, and enhance scheduling efficiency. According to experimental results, SVM and Neural Network based models has the highest accuracy, which makes it a promising method for intelligent appointment scheduling in healthcare.

An AI-Based Hands-Free Communication System Using Eye Blinks and Morse Code with YOLOv8 and LSTM Prediction

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Abstract: Individuals with severe motor and speech impairments face significant barriers in expressing their needs and thoughts through conventional methods of communication. This research proposes an eye blink-and Morse code-based hands-free assistive communication system. The proposed system uses a camera and takes advantage of an effective deep learning model inspired by YOLOv8, which can detect eye states and classify blink durations into short and long blinks with high accuracy in real time. These will be mapped to Morse code dot and dash and decoded to readable text. An AI-based next-word prediction mechanism and some predefined shortcut commands for emergency situations have been integrated to improve efficiency in communication. This system also supports the Hear Me feature, enabling audio communication through text-to-speech output. Experimental evaluation has shown that the proposed system achieves 98.9% for recognition of blinks, showing its improved performance against other state-of-the-art approaches. It proves that the system is reliable and efficient and well-suited for real-time assistive communication, hence offering a more accessible intelligent solution for users with severe physical disabilities.

Deep Learning-Based Early Detection of Dysgraphia in Children Using EfficientNetB0 Architecture

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Abstract: Dysgraphia, a learning disability affecting handwriting fluency and written expression, often remains undiagnosed in early childhood, leading to significant academic and emotional challenges. Traditional diagnostic methods are time-consuming, subjective, and require specialist intervention. This paper presents an automated deep learning-based system for early detection of dysgraphia using handwriting analysis. We developed a convolutional neural network (CNN) model based on the EfficientNetB0 architecture to classify handwriting samples into three severity levels: Normal, Low Dysgraphia, and High Dysgraphia. The model was trained on a comprehensive dataset of 105,000 handwriting images compiled from multiple sources. Our approach achieved an accuracy of 96.32% with an F1-score of 0.96, demonstrating robust performance in identifying handwriting anomalies. The system was integrated into a web-based application using Flask, enabling real-time screening accessible to teachers, parents, and clinicians. This work contributes to inclusive education by providing an efficient, scalable, and accessible tool for early identification and intervention of dysgraphia in children.

AI-Powered Smart Rural Health Monitoring and Water Quality Prediction System

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Abstract: The ability to obtain safe and drinking water is a significant global responsibility, especially with advancing technologies, such as artificial intelligence which expands accessibility in water quality evaluation. The proposed system utilizes machine learning methodologies for predicting water potability based upon various testing pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity are examples of physicochemical parameters. The dataset was cleaned and standardized for accuracy using Python prior to training with the model. Seven algorithms (XGBoost, Random Forest, LightGBM, SVM, Neural Network, KNN and Logistic Regression) were studied, with XGBoost performing the best in regard to accuracy and reliability. In order to add transparency with the intelligent model, To explain how each parameter contributes to the predictions, Explainable AI (XAI) tools like SHAP and LIME were added. A user-friendly Streamlit web application was developed that allows users to input data or to upload files for a potability evaluation instantly along with connected disease risk. Overall, the intelligent system presents an easily replicable and trustable solution for monitoring water quality in order to help public health and environmental sustainability efforts.

AI-Driven Integration of Imaging, Genetic, and Epigenetic Signals for Alzheimer's Disease Diagnosis

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Abstract: Accurate and early prediction of Alzheimer's disease (AD) is vital in neurodegenerative research. This review examines 10 key studies from 2022 to 2025, focusing on recent advances in artificial intelligence that allow for the integration of different types of biomarkers, including magnetic resonance imaging (MRI), peripheral blood DNA methylation, genetic variants, and clinical data. These methods offer minimally invasive options for diagnosis and prognosis and they could be used in smart city healthcare systems to provide scalable, preventive diagnostics for aging urban populations. The review highlights deep learning models that utilize longitudinal data, attention mechanisms, and multimodal fusion. CNN-Transformer hybrids and cross-attention frameworks outperform single-modal models by improving sensitivity and interpretability across various modalities. The integration of structural MRI with methylation and genomic profiles enables the development of scalable, biologically grounded models for early detection of Alzheimer's disease. Key challenges remain, such as limited cohort diversity, computational complexity, sustainability issues like energy efficiency of large models, fairness in multi-ethnic validation, and the need for real-world validation. Future directions include federated learning for collaborative urban health networks, self-supervised multimodal pretraining, causal modeling, uncertainty-aware predictions, and efficient AI deployment for clinical use in smart hospitals and IoT-integrated systems.

A Vessel-Aware Preprocessing Framework with Deep Neural Networks for Accurate Retinopathy of Prematurity Classification

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Abstract: Retinopathy of Prematurity (ROP) is a major reason of avoidable blindness in newborns. Appropriate and accurate identification of the disease and classification and its stages on time is critical in the diagnosis and treatment. An automated ROP retinal image analysis procedure with the help of advancements in deep learning improves the identification and grading of the disease. Deep learning-based approaches in various stages of the procedure improves image quality and preserve the significant vascular structural details of the retinal image. This paper proposes a vessel-aware preprocessing framework integrated with deep neural networks for high accuracy classification of ROP. The proposed methodology has used the median filtering, noise cancelling, localized adaptive histogram equalization, contrast enhancement and a structure-preserving Non-Local means denoising strategy guided by vessel mask with good vascular feature preservation while suppress background noise. Augmentation is used to increase strength and generalization. The enhanced retinal fundus images then classify using the assistance of a deep neural network-based model, perform stage wise ROP categorization according to the ICROP3 protocol. Experimental evaluation proves that the proposed vessel-aware preprocessing methodology enhances classification performance by achieving an overall accuracy of 92.73%, sensitivity of 93.52%, specificity of 92.10% and an area under the ROC curve (AUC) of 0.96. The stage wise analysis also proves a consistent and reliable performance in early and advanced stages of ROP which include the Plus disease. The findings indicate that vessel-aware preprocessing of images enhances the deep learning-based ROP screening and classification significantly.

Machine Learning Approach For Rapid ECG-Based Cardiac Anomaly Detection

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Abstract: Heart disease is one of the major problem in the world for death rate increasing. Patients do not show clear symptoms until the condition becomes serious. Because of this, early detection of heart problems is very important. In traditional ECG analysis usually requires experienced doctors and may take more time. In this project, we mainly focused on building a machine learning based application to detect heart related problems from ECG signals. We used in this project XGBoost algorithm to classify ECG data into seven different categories such as normal, atrial fibrillation, tachycardia, bradycardia, arrhythmia, other conditions, and noisy signals. We trained the XGBoost algorithm using more than 40,000 ECG samples. In was collected from different hospitals, which helped improve the accuracy of the data. At the time of testing, we observed that the system gives good classification results and produce results in real time. In that we observe, this project may be useful for the early detection of heart disease and may help for medical professionals during diagnosis.

AI-Based Genomic Risk Stratification and Drug Response Prediction: A Multi-Modal Deep Learning Framework

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Abstract: Precision medicine relies on predicting genetic disease risk and patient drug response. Conventional methods are challenged to combine diverse types of genomic information and handle population diversity. This paper presents a wide artificial intelligence framework to combine clinical variant databases (ClinVar), genome-wide association studies (GWAS), patient genotypes (1000 Genome Project), and pharmacogenomics (PharmGKB). This framework analyzes 58,916 genetic variants on even-numbered chromosomes from around 1,000 patients. Sophisticated preprocessing techniques and intelligent cross-reference imputation techniques achieve 98.2% data completeness. New feature engineering techniques extract 116 genomic risk features, including Variant Impact Composite Scores and Gene Burden Scores. BioBERT embeddings extract semantic relationships between 324 categorical clinical annotations. For drug response, 16,038 pharmacogenomic associations involving 1,068 drugs and 1,195 genes are analyzed. Specialized techniques for metabolizer phenotype prediction are also included. The genetic risk model employing a transformer model achieves 83.05% accuracy and 98% precision on high-risk variants among 3.4 million patient samples. Similarly, the deep neural network-based drug response model obtains 97.92% accuracy and 88.52% recall on toxic reactions despite a 2.53% class imbalance. Generalization capability is demonstrated with validation and test set gaps of less than 1%. This study presents research-grade results as a proof of concept for precision medicine.

Estrogen Endocrine Disruptors-Driven Cervical Cancer Subtypes: Unraveling Heterogeneity Through Multi-Omics Integration of Estrogen and Immune Profiles

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Abstract: Cervical cancer remains a major and increasing burden in many low- and middle-income countries, where advanced disease is often highly aggressive, metastatic, and frequently refractory to standard therapies. However, it is still unclear how estrogen-targeting endocrine disruptors (EEDs) mainly bisphenol, phthalates and dioxin shape cervical cancer heterogeneity and immune responsiveness. This study explores the role of estrogen endocrine disruptors (EEDs) in cervical cancer by jointly acting on estrogen signaling and tumor immune microenvironment. In the current work, these EED activities are quantified by ssGSEA-derived scores, and a consensus clustering is formulated to identify three stable patient clusters. DIABLO multi-omics integration confirmed that the three clusters possess distinct molecular features, such that Naive Bayes classifiers accurately predicted cluster membership from compact LASSO-selected multi-omics & mRNA signatures. Cluster 1 showed high EED/high TIDE scores, whereas Cluster 3 had low EED/low TIDE, indicating greater predicted immunotherapy sensitivity. Further the gene-regulatory network analyses demonstrated hub regulators TFDP1 in linking EED exposure to immuno-endocrine phenotypes, nominating the EED–estrogen–immune axis as a potential biomarker and therapeutic target in cervical cancer.

Pan-Cancer Analysis for the Identification of Common Oncogenic Biomarkers Using Platelet RNA-Sequencing Data

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Abstract: Cancer diagnosis and prognosis increasingly rely on molecular biomarkers that reflect shared oncogenic pathways across tumor types. In this study, a pan-cancer transcriptomic analysis was performed using RNA-sequencing data from blood platelets to identify common oncogenic biomarkers across breast, colorectal, glioblastoma, hepatobiliary, non-small cell lung and pancreatic cancers. Global expression patterns were assessed after batch correction followed by differential expression analysis with edgeR. A total of 407 common DEGs were subjected to pathway enrichment analyses using GSEA, Gene Ontology, Reactome, Hallmark, KEGG and WikiPathways databases. Protein–protein interaction, expression-to-kinase interaction and transcription factor enrichment analyses were conducted to identify regulatory hubs. Breast and lung cancers showed predominantly downregulated genes whereas glioblastoma, pancreatic and colorectal cancers showed upregulated genes while hepatobiliary cancer exhibits a relatively balanced distribution. Enrichment analyses consistently revealed platelet-associated pathways including platelet activation, thrombin signaling, coagulation and calcium-dependent signaling, along with metabolic pathways such as glycolysis and the pentose phosphate pathway. In contrast, ribosomal biogenesis, translation and immune-related pathways were significantly downregulated with the strongest suppression observed in glioblastoma. Reactome pathways showed the highest enrichment significance among all databases. Network analyses suggested that transcription factors such as MAX, CREB1, NFYB, NFYA, TAF1, UBTF, YY1, GABPA, and ELF1 may function as central regulatory hubs commonly observed across multiple cancer types. Overall, this study highlights platelet-driven signaling, calcium homeostasis, coagulation and transcriptional regulation as core pan-cancer features supporting the potential of platelet-derived transcriptomic biomarkers for non-invasive cancer diagnosis.



Track 2

Smart Mobility and Intelligent Transport Systems

Rail Rover: Integrated Rail Track Cleaning And Crack Detection System

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Abstract : Today global transportation highly relies on the railway track network, hence there is a need for maintaining the cleanliness of the railway network with at most structural efficiency and safety. Conventionally, the maintenance technologies appear to be risky and labor-intensive increasing the overall expenditure of the system. The Rail Rover system is introduced as an autonomous system that perfectly integrates the rail track cleaning and the crack identification systems with the help of machine learning techniques. The Rail rover's high-resolution cameras and the ultrasonic sensors that assist in the identification and classification of the waste materials on the tracks. Together with this, it continuously monitors the rails for the presence of any structural irregularities and notifies the same to the concerned crews in real time for establishing prompt actions, that enhances the security level of the entire system. In this way, the Rail Rover system accomplishes its image processing and machine learning tasks. Through these procedures, the Rail Rover system has been observed to eliminate the need for manual labor, it saves the operating expenses, and offers an improvised waste management strategy. It is a useful instrument for contemporary railway maintenance and greatly increases the efficiency of railway systems.

Comprehensive Analysis Of Path Planning Algorithms For Autonomous Navigation Of Humanoid Robots

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Abstract: Intelligent and efficient path planning is crucial for humanoid robots to autonomously navigate challenging and unpredictable circumstances. To navigate swiftly and securely these robots must constantly adjust to obstacles, shifting terrain, and real-time environmental changes. Due to their deterministic nature and comprehensible behavior, classical algorithms such as the A* method, Dijkstra's algorithm, Voronoi diagram based navigation, and the artificial potential field technique have been widely used. Nevertheless, they have drawbacks such as poor scalability in large spaces, increased non-linearity in complex navigation tasks, and high processing demand for real-time humanoid execution. Learning-based solutions with more adaptable decision-making, faster processing, and improved generalization across untested scenarios have been made possible by recent advancements in artificial intelligence, especially large language model-based reasoning systems. These learning based planners make better use of large sensor datasets and semantic context to reason about navigation, dynamic obstacles, and environmental unpredictability. The artificial potential field technique, Voronoi based planning, A*, Dijkstra's algorithm, and large language model-based reasoning for path planning for autonomous humanoid navigation are all compared in this research. To make sure these algorithms are appropriate, they are tested under identical experimental settings. Strengths, weaknesses, and situational performance are emphasized in the comparison. The trade-offs for each strategy are dependent on execution behavior and real-time limitations.

Edge-Accelerated Hardware-Efficient Image Enhancement for Intelligent Traffic Surveillance

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Abstract: Rapid urban growth has led to severe traffic congestion, frequent rule violations, and growing safety concerns, demanding more intelligent and automated surveillance systems. Traditional monitoring solutions that depend on manual observation or high-cost GPU servers lack the flexibility, scalability, and power efficiency required for real-time deployment. To overcome these limitations, this work proposes an edge-based smart traffic surveillance framework built on FPGA hardware, enabling efficient, low-latency image enhancement directly at the source. The system is designed to integrate traditional enhancement algorithms with lightweight AI models, ensuring both visual clarity and adaptability under diverse environmental conditions. Its flexible architecture allows customization of convolutional neural networks according to hardware constraints, achieving an optimal balance between performance and power consumption. By comparing classical and AI-driven approaches in terms of processing speed, hardware utilization, and enhancement accuracy, the proposed design establishes a scalable and energy-efficient foundation for next-generation intelligent traffic surveillance applications.

Smart Transit Guardian: Seamlessly Navigating The Intersection Of Safety & Efficiency

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Abstract : The lack of real-time information regarding emergency vehicle delays at railway crossings and road construction sites is an ongoing issue, due to the inability of traditional GPS and traffic applications to navigate around these temporary obstructions. This paper describes a new IoT-based solution called Smart Transit Guardian, which allows emergency vehicles to use ESP32-based infrared proximity sensors at railway crossings along with the Google Maps API for up-to-the-minute route recalculation (within 1 second). The architecture is based on Firebase cloud services and provides two distinct user interfaces: 1) for emergency responders and 2) for the general public, which allows for two-way communication between the two user groups for accident reporting and monitoring of railway gate status as well as dynamic optimization of routes. Testing has shown that average emergency response times have decreased from 10-15 minutes to 5-7 minutes as well as a significant increase in avoidance of closing gates while responding (from 40-50% avoiding to 90%) with a route data refresh interval of every 5 seconds and an average processing delay of 0.2 seconds per refresh; thus making it a practical, deployable solution for smart city emergency management.

AI Driven Driver Safety And Fatigue Monitoring System

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Abstract: Driver fatigue significantly increases the risk of road accidents due to reduced attention, delayed reaction time, and impaired decision-making. This paper presents an AI-driven driver safety and fatigue monitoring system designed to detect early signs of drowsiness using real-time computer vision and multimodal interaction. The system analyzes facial landmarks to compute Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) for identifying prolonged eye closure and repetitive yawning patterns. Upon detecting fatigue indicators, a progressive alert mechanism consisting of visual, auditory, and voice-based confirmation is activated. If the driver fails to respond, an automated emergency protocol transmits geolocation information to predefined contacts. The system incorporates adaptive thresholding and real-time processing to maintain reliability under varying environmental conditions.

Comparative Analysis of Classical Guidance Laws for UAV Navigation

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Abstract: Guidance laws enable UAVs to operate independently by computing the required flight path from the current position to the target, making them well suited for autonomous missions, long duration flights, and operation in dynamic environments. They also enhance mission reliability and safety by enabling accurate path tracking, obstacle avoidance, and robust performance under uncertainties. This paper focuses on four of the most widely adopted techniques: proportional navigation, pure pursuit, waypoint navigation, and line of sight, each of which adopts a distinct guidance law for path control. The main objective is to explore the tracking performance, stability, and effectiveness of these algorithms in UAV navigation. All techniques were evaluated in a MATLAB-based simulation environment, and a comparative analysis was performed in terms of tracking error and computational cost. The individual performance of the guidance algorithms is evaluated, and their behavior is analyzed in a two-dimensional plane. A comparative analysis of these guidance algorithms is carried out using spline trajectories, straight-line trajectory, and S-curve routes, for which the tracking error performances are examined

Multi-Object Detection in Autonomous Vehicles Under Adverse Weather Conditions Using Modified RT-DETR

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Abstract: Environmental perception is one of the important features of Autonomous vehicles (AV). Autonomous vehicles use deep neural networks (DNNs) for accurate object detection in various environments. However, adverse weather conditions like rain, fog, and snow significantly degrade detection performance, leading to potential safety risks. This work proposes a modified RT-DETR (Real-Time Detection Transformer) architecture for multi object detection in adverse weather conditions. Here the standard ConvNeXt backbone of RT-DETR is replaced with the YOLOv11 Medium backbone. This modification significantly enhances the model's feature extraction capability and inference speed while preserving the benefits of transformer-based attention mechanisms. The proposed architecture is evaluated alongside YOLOv11 variants (Nano, Medium, Large) and the original RT-DETR using images from the DAWN dataset, which contains diverse real-world scenes captured under fog, rain, and low-light conditions. Experimental results demonstrate that the modified RT-DETR outperforms all baseline models in terms of precision, real-time latency, and robustness in complex, weather affected visual environments. Our experimental results demonstrate improved detection robustness compared to conventional models, with a 15% increase in precision and a 20% reduction in false negatives. The proposed system enhances safety in autonomous driving and real-time surveillance applications.

Dynamic Time Handling for Traffic Optimization Using Machine Learning and SUMO Simulation

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Abstract: The rush of urban traffic is still a frequent question in rapidly growing cities, which contributes to increased delays, fuel consumption, and environmental pollution. Traditional traffic signal systems with a fixed time have a lack of flexibility to respond to dynamic and unexpected traffic conditions, which often leads to disabilities such as increased congestion and longer wait times for vehicles and pedestrians. This research proposes a dynamic traffic signal optimization framework run by the machine learning model to predict optimal green light duration at a four-way intersection. Several algorithms, including polynomial regression, random forest regression, XGBoost, and multi-layer perceptron (MLP), were implemented, and their prediction was compared to Indian cities for accuracy using a synthetic traffic dataset that represents different traffic scenarios from Indian cities. Of these, the random forest achieved the highest performance, an average directional accuracy of 96.67% and a total accuracy of 87.23%. The estimated signal reduction of the model was integrated into the Sumo (Simulation of Urban Mobility) platform for simulation-based verification. The results indicated that the dynamic, machine-driven system made the traditional fixed-early schemes significantly improved by reducing the vehicle's passive time and delaying intersection and overload levels. This study emphasizes the ability of AI-Mangoing adaptive traffic control systems as scalable, efficient, and practical solutions to increase urban dynamics in the environment of the smart city.

Real-Time Analytic Approach To Finding Nearby Transportation Using Clustering Algorithm

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Abstract: Transportation is an important part of our everyday life, and the ability to find the closest available service as quickly as possible is still a major issue. This paper aims to come up with a real time analysis method to find nearby transportation services with the help of clustering algorithms. This system gathers location-related information and processes it to cluster transportation points in space, therefore, allowing users to find the most useful nearby ones quite easily. The proposed system minimizes the search space, increases the accuracy and responsiveness of search results by using clustering techniques. The rural livelihoods predominantly depend on agriculture; hence access to modern farm machinery like harvester machines is imperative in enhancing better agricultural productivity in the rural areas and reduce losses that are experienced after harvesting the crop. In solving the problem, the proposed system will be expanded within the framework of the real-time analytical model to cover transportation services as well as harvester machines. The system gathers the geolocation data of farmers and of service providers and runs clustering algorithms to cluster machines according to their proximity and provides the users with the optimum recommendations. The suggested solution will help urban commuters and the rural farmers with the recommendation accuracy significantly by optimally structuring spatial data and minimizing search time. The available transportation services like taxis and buses in the urban areas are usually geographically spread out causing ineffective manual searches and delays in travel. The current applications often make users choose too much and irrelevant without properly reflecting on real-time proximity and availability. The suggested clustering-based solution helps overcome these constraints by placing particular emphasis on the geographically relevant and available resources.

Enhancing UAV Image Object Detection Using light weight Attention Modules In YOLOv5

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Abstract: Unmanned Aerial Vehicles (UAVs) are increasingly utilized in applications such as surveillance, traffic monitoring, agriculture, and disaster management. However, object detection in aerial imagery remains challenging due to small object sizes, dense distributions, and complex backgrounds. To address these challenges, this work presents an enhanced YOLOv5-based object detection framework by integrating Positional Enhancement Attention (PEA) and Adaptive Positioning Deformable Convolution (APDC). The PEA module strengthens spatial feature representation by emphasizing important regions, while the APDC module improves localization by dynamically adjusting convolutional sampling positions based on object structure. The proposed approach maintains the real-time efficiency of YOLOv5 while improving detection performance. Experimental evaluation on the VisDrone dataset demonstrates improved mean Average Precision (mAP) compared to the baseline model, showing the effectiveness of the proposed enhancements.

Real-Time Obstacle Detection And Risk Assessment For Autonomous Vehicles: An Ensemble With Adaptive Low-Light Enhancement

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Abstract: Ensuring safety in real-world driving and autonomous vehicle environments requires intelligent systems capable of detecting obstacles early and providing timely warnings. The existing sensor solutions which use LiDAR and RADAR technology provide high precision but their costs make them unfeasible for both consumer products and autonomous vehicle systems. This project presents a lightweight, laptop-deployable ensemble deep learning system for real-time obstacle detection, distance estimation, and intuitive risk-level alerts for intelligent driving systems. The detection pipeline uses YOLOv8n and a lightweight segmentation model (FastSAM) to create an output which uses IoU-based fusion for improved strength. The system uses Zero-DCE++ to improve visibility in dark environments by providing low-light image restoration which achieves accurate night-time detection without needing extra equipment. The distance estimation module uses camera geometry to determine obstacle distance, which enables the system to identify three different status levels: safe, caution and critical. The complete system functions in real-time through a standard laptop which uses a dashcam, making it an affordable solution for both academic research and autonomous vehicle testing. Experimental evaluations on custom driving footage and benchmark datasets demonstrate that the proposed ensemble achieves improved detection consistency, reliable distance estimation, and rapid risk signaling suitable for practical autonomous and driver-assistance systems.

A Novel Framework For New York Taxi Trip Data Pattern Identification Using Big Data Analysis

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Abstract: Taxis are a vital mode of transportation in New York City, generating a wealth of data with every trip. Our aim was to utilise this data, analyzing records from NYC taxis to reveal meaningful insights. The ultimate goal is to help taxi services move beyond just reacting to demand and start proactively providing more efficient, responsive service to their users. However, the sheer volume and velocity of this data present a significant engineering challenge that traditional systems simply cannot handle. To tackle this, we designed and implemented a scalable, modern data stack. We process raw trip data sourced directly from the NYC Taxi and Limousine Commission (TLC) using a robust ETL (Extract, Transform, Load) pipeline. This pipeline is built and orchestrated with the open-source tool Mageai, is responsible for the crucial work of cleaning, transforming, and preparing the data for analysis. For storage, we employed a two-tiered cloud-native approach: Google Cloud Storage (GCS) serves as our flexible data lake for ingesting the vast amounts of raw data, while Google BigQuery acts as our high-performance data warehouse, enabling rapid, complex querying. This powerful setup allowed us to analyze the data at scale, revealing several key patterns. Our analysis confirmed that credit cards are, by a large margin, the most preferred payment method. More critically, we quantified the significant, persistent demand for rides to and from high-traffic hubs like JFK International Airport—a vital insight for optimizing vehicle dispatch. The data also shows us interesting human behaviors, such as pinpointing specific locations where riders frequently negotiate fares. These findings are more than just statistics; they provide actionable intelligence, helping stakeholders make smarter, data-driven decisions for a better taxi service.

V2V-LLM: Large Language Models For Cooperative Driving

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Abstract: Effective cooperation between autonomous vehicles requires more than sensing and obstacle avoidance, vehicles must also communicate their intentions and reason about shared traffic situations. Most current cooperative driving strategies relies on rigid rule sets or predefined algorithms, which makes them less responsive in dynamic environments. This project presents V2VLLM, a cooperative driving framework where autonomous vehicles exchange information through natural-language messages and rely on a Large Language Model (LLM) to perceive the situation and decide the next action. Each vehicle disseminates its status and intended action, and the LLM evaluates all received messages to select a suitable high-level command, such as yield, keep lane, or adjust velocity. The selected action is then translated in to motor control signals within the Webots simulation environment, enabling real-time vehicle behavior. A memory component records previous interactions and responses, allowing the system to improve decision-making in future situations. Simulations conducted in Webots across different traffic configurations confirm that V2V-LLM facilitates cooperation and reduce conflicts compared to deterministic rule-based strategies. The findings suggest that natural-language-enabled reasoning can be a realistic direction for improving autonomous vehicle coordination.

Rolling Shutter Correction And Temporal Super-Resolution: Evolution, Limitations, And Emerging Directions

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Abstract: Rolling shutter (RS) sensors, common in mobiles and embedded cameras, capture images via row-wise readout and therefore introduce motion-dependent geometric distortions under camera or scene motion. Correcting these artifacts and recovering distortion-free global shutter (GS) frames—or temporally consistent GS video—is critical for downstream tasks such as visual odometry, 3D reconstruction, and autonomous perception, particularly in smart mobility and intelligent transport systems where cameras are widely deployed in traffic surveillance, connected the rapid evolution of rolling shutter correction (RSC) and rolling shutter temporal super-resolution (RSSR), from early analytic geometry-driven methods to recent data-driven, self-supervised, and generative approaches. The survey categorizes and analyzes representative works across the following paradigms: classical geometric and optimization-based models, supervised deep learning frameworks, self-supervised and event-guided methods, and generative and diffusion-based techniques. The survey aims to provide a structured reference that clarifies current strengths and limitations, and to guide future research toward robust, efficient, and generalizable RS correction systems suitable for real-world intelligent transportation and mobility-centric vision applications.

AI-Powered Smart Driving Assistant System

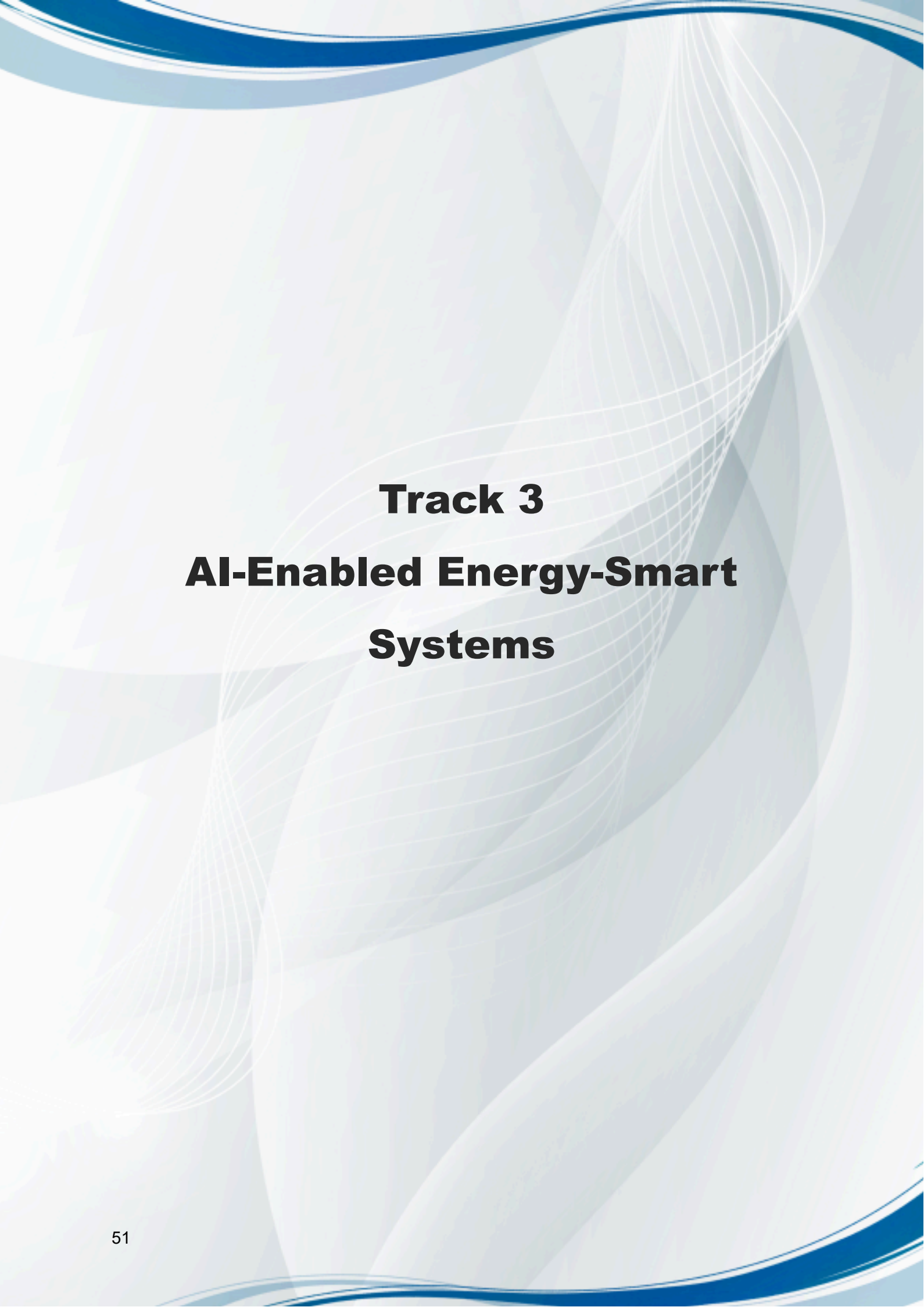
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Abstract: Road accidents in India frequently occur due to overspeeding, poor traffic sign visibility, and driver distraction, highlighting the need for affordable real-time driver assistance systems. This paper presents an AI-Powered Smart Driving Assistance System that provides low-cost, real-time road monitoring using computer vision and embedded processing. The system is implemented and experimentally validated on a Raspberry Pi 5 platform integrated with a camera module and an RPM-based vehicle speed sensing unit. A Python Flask-based simulation backend was initially developed to emulate hardware functionalities, enabling controlled testing and validation prior to full system deployment. A Flutter-based mobile application displays live video frames, AI-generated traffic sign detections, real-time vehicle speed, and overspeed warning alerts, while also maintaining violation logs and generating a driving performance score. The proposed system performs real-time traffic sign recognition and context-aware speed monitoring entirely on-device, ensuring low latency and data privacy. Experimental evaluation demonstrates reliable detection performance and stable real-time operation, making the solution a scalable and cost-effective alternative to conventional ADAS technologies.

Machine Learning Based Drone System For Water Detection And Monitoring

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Abstract: This paper presents how a drone-based system is useful for non-contact water flow detection. Also, it is able to differentiate between land and water, also detects the water flow speed and the depth. The proposed system integrates an ESP32-CAM module fronting downward to capture aerial footage. This vision-based method abolishes the need of physical contact with water and makes the system perfect for applications in floods monitoring, agriculture, and remote surveying.



Track 3

AI-Enabled Energy-Smart Systems

A Fully Supervised Framework for Lightweight Soiling Segmentation on Photovoltaic Panels using YOLOv8

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Abstract: Photovoltaic (PV) solar panels gather soiling, the build-up of dust and dirt from the environment. For efficient automated maintenance, a system must detect not only soiling but also the intelligent timed maintenance. This paper describes an end-to-end intelligent soiling monitoring fully supervised system and alert generation. The light-weight YOLOv8n-seg model, which accurately performs pixel-level segmentation, is the system's fusion vision quadrant, which integrates to a bigger system of intelligent decision-making based on local weather forecasts, suppressing cleaning alerts when rain is predicted to save resources. Trained and validated on the Translucent which is a collection of over 40,000 images with corresponding segmentation masks our model puts forth very robust results. We have put together a base which is at the same time practical and resource efficient for fully automatic solar panel maintenance systems. Also we have designed a web based dashboard for real time panel status visualization and a database for past soiling data.

Hybrid GA–PSO Optimized PID Control for Frequency Stability Enhancement in Inverter-Dominated Microgrids

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Abstract: Inverter-dominated microgrids face significant frequency stability challenges due to reduced physical inertia and the variability of renewable energy sources. This paper presents a Hybrid Genetic Algorithm–Particle Swarm Optimization(GA–PSO) approach for tuning PID controllers used in secondary frequency regulation. A reduced-order surrogate twin model is developed to represent the key dynamic behavior of the microgrid while keeping the computational requirement suitable for real-time implementation. The controller parameters are optimized using a multi-objective cost function that considers peak frequency deviation, settling time, integral error, and control effort. The proposed method is evaluated in MATLAB/Simulink under different disturbance scenarios, including photovoltaic power variations, load changes, and generator outages. Simulation results indicate that the Hybrid GA–PSO tuned controller achieves lower transient deviation and faster settling compared to GA, PSO, and ACO based tuning methods.

Optimal Dispatch of Hydrogen Supported Integrated Energy Systems via an Adaptive Mountain Gazelle Optimizer

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Abstract: The increasing penetration of renewable energy sources (RES) undergoes major changes into smart power systems that demands effective coordination among electricity and hydrogen based storage. An integrated energy system (IES) with hydrogen conversion provides a better solution, but their operation needs advanced optimization models because of high non-linearity and multi-energy coupling. This work suggests an optimized operational strategy for an IES incorporating wind, photovoltaic generation, electrolytic hydrogen production, fuel cells, and hydrogen storage. For improving economic performance and reducing carbon emissions, a multi-objective scheduling model is defined and solved by an Adaptive Mountain Gazelle Optimization (AMGO) algorithm. A Lévy-mutation operator is combined for strengthening global exploration and avoiding early convergence. This provides efficient handling of the nonlinear characteristics of IES optimization. The suggested approach optimizes power dispatch, hydrogen production, and storage utilization under varying renewable generation conditions. Experimental outcomes show that the AMGO achieves lower operating cost of 35.2% and CO₂ emission reduction 95.3%. The analysis concludes the potential of AMGO for managing hydrogen supported low carbon energy systems

Machine Learning Based Smart Grid Energy Optimization on Embedded Edge Systems

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Abstract: This paper presents a comprehensive architecture for smart grid energy optimization that tightly integrates low-power edge computing with machine learning (ML) models, advanced sensing, and secure trust mechanisms. We propose an embedded edge computing layer featuring ultra-low-power ARM/RISC-V processors with on-chip ML accelerators (e.g. Google Coral EdgeTPU at ~ 2 TOPS/W) that perform real-time inference on demand-response policies and load forecasts. A smart grid sensing layer (with smart meters, grid sensors, and renewable monitors) feeds time-series data into the system. The ML core executes short-term/long-term load forecasting (LSTM, Transformer, XGBoost), RL-based energy dispatch, and predictive fault detection, all on-device or in a hybrid edge-cloud fashion. We also incorporate federated learning across substations and confidence-aware decision-making. A secure data/trust layer uses blockchain for immutable transaction logging and smart contracts to enforce demand-response policies, along with zero-trust networking and hardware TEEs for device authentication. Experimental validation (on a Google Coral prototype in an Indian smart city grid) demonstrates edge vs. cloud latency (~ 10 ms vs. ~ 100 ms), power usage (≈ 500 mW vs. 5 W), and high accuracy (≈ 93 – 95% forecast accuracy, 95% fault-detection precision). Our results show substantial gains in throughput and energy efficiency under city-scale loads. The proposed system advances secure, real-time ML-driven smart grid control on embedded platforms.

AI-Powered Drone System for Smart Inspection and Fault Detection in Power Distribution Lines

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Abstract: The reliability of electrical power distribution is essential for modern infrastructure, yet traditional manual inspection methods are often time-consuming, hazardous, and inefficient. This paper presents an AI-powered drone system designed for smart inspection and fault detection in power distribution lines. The proposed system integrates Artificial Intelligence (AI), Unmanned Aerial Vehicles (drones), and the Internet of Things (IoT) to provide a safer, faster, and more accurate method for identifying faults. Drones equipped with cameras capture live images and electrical parameters from power lines, which are analysed using advanced AI algorithms such as YOLO (You Only Look Once) to detect defects like damaged insulators, conductor breakage, corrosion, and overheating. Meanwhile, IoT-enabled nodes continuously monitor voltage, current, and temperature data, transmitting real-time information to a web interface. The integration of these technologies allows instant fault localization, automated reporting, and proactive maintenance, thereby reducing human risk and minimizing power outages. The system supports the goals of sustainable energy management by improving grid reliability, enhancing operational efficiency, and reducing the environmental impact of traditional inspection practices. Overall, this paper demonstrates a cost-effective and innovative approach toward achieving smart grid modernizations and contributes to several UN Sustainable Development Goals (SDGs), including affordable energy, innovation, and climate action.

Bi-Directional Power Flow Control in Grid Interfaced Solar Water Pumping With PWM_on_PWM BLDC Motor Drive

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Abstract: This article presents an advanced methodology for enhancing motor torque performance within grid-tied solar photovoltaic (PV)-fed water pumping system using Brushless DC motor. The designed water pumping system integrates a brushless DC (BLDC) motor drive which can triumph over a traditional water pumping system in various features. Unlike standalone systems, utilization of solar power along with grid interfacing ensures continuous operation by draining power from the grid during varying weather conditions. Grid-connected setups can transfer excess solar energy to grid thereby reducing energy wastage and can be utilized as a source of income. This attribute enhances the overall reliability of the proposed system, confirming the continuous utilization of the water pump at its maximum capacity. Torque ripple generation is a major drawback affected by usage of BLDC motor pump and in order to overcome this issue an innovated PWM switching scheme is implemented in-order to minimize the accumulation of torque ripples. The implementation of bidirectional power flow control is accomplished using a single-phase voltage source converter (VSC). This VSC utilizes a unit vector template generation method, which effectively regulates the exchange of power between the power grid and the Voltage source inverter's (VSI) DC bus. The proposed system enhances performance by reducing electromagnetic torque disturbances, improving power quality, and ensuring better motor control and exhibits a comparison of the total harmonic distortion (THD) of the system when operating solely on the utility grid versus when both solar and the utility grid are being used.

AI-Enabled Smart Border Surveillance System with Multi-Sensor Fusion and Embedded Threat Detection

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Abstract: Border security is a critical pillar of national defense, demanding persistent monitoring and rapid threat neutralization. Traditional surveillance, often reliant on manual observation and single-sensor setups, fails under challenging environmental conditions such as low-light, fog and complex terrain. To overcome these limitations, this paper presents an AI-enabled smart border surveillance system built on an intelligent multisensor fusion framework. The system synergistically integrates a VL53L0X LiDAR for precise range mapping, an RCWL-0516 microwave radar for motion detection, an MLX90640 thermal array for heat signature analysis, and an ESP32-CAM for visual verification. A Raspberry Pi Zero 2 W serves as the central embedded computing unit, leveraging TensorFlow Lite for ondevice processing. By deploying an optimized, INT8-quantized YOLOv11n (Nano) model, the system performs real-time classification of humans, vehicles, and threats locally, entirely eliminating cloud dependency. Simulation and hardware validation demonstrate strong performance in challenging environments where traditional systems fail. The system achieves high detection accuracy (mAP > 90% for people and vehicles) with an end-to-end latency of under 2.0 seconds. This work provides a scalable, low-power, and self-sufficient solution for next-generation border security by showcasing the real-world integration of advanced AI and multi-sensor fusion at the edge.

DQN based EV Charging Scheduling for Distribution Grid Load Balancing

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Abstract: The increase in adoption of Electric vehicles creating significant stress on distribution level power grids due to uncoordinated charging and unpredictable peak energy demand. Traditional scheduling methods like fixed time allocation or static loads management rely on predefined rules and historical averages rather than real-time system conditions. These approach assumes predictable demand and stable grid behaviour, making them unable to adapt to the real time grid fluctuations or varying user requirements. This paper proposes an intelligent EV-charging station scheduling system based on Deep-Q-Network that learns optimal decisions from grid state transitions modelled through Markov Decision Process. The system collects user requests through a mobile application and process them on the central server. The server has the Realtime information of the grid load, station availability, and booking status. While considering the user information's and the grid conditions the DQN agent identifies the top two or three most suitable charging stations. Priority rules are built in to the system to make sure the emergency vehicles such as ambulances, fire trucks and public transport vehicles receives priority in charging. This balances the travel distance, station load and available capacity across the distribution grid. Once the station selected by the user the system confirms the booking that reserves the slot. At the charging station a ML based licence-plate verification mechanism is used to authenticate the arriving vehicle against the server side booking record. This ensures that only authorized vehicle can charge from the reserved slot. Experimental evaluation demonstrated that the proposed approach effectively balances the charging demand across the grids, reduces the stress on the grid and improves user waiting times. Overall, the system provides a scalable and automated solution for EV charging and supports sustainable integration of EVs in to existing power infrastructure.

Synergistic Solar Photovoltaic and Vertical Axis Wind Turbine Generation

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Abstract: This paper presents the design and simulation of a hybrid renewable energy system integrating solar photovoltaic (PV) panels and a Vertical Axis Wind Turbine (VAWT) specifically designed for installation along highways. The system aims to harness the dual renewable resources of solar irradiance and wind energy, including the kinetic energy from passing vehicles, to provide a sustainable and decentralized power source. Key components of the system include a PV array with a Maximum Power Point Tracking (MPPT) controller, a Permanent Magnet Synchronous Generator (PMSG) driven by a VAWT, a battery energy storage system (BESS), and a grid-tied inverter. The system's architecture is designed to ensure continuous power generation by leveraging the complementary nature of solar and wind resources. This paper details the modeling of individual components and the integration of the complete system, followed by a simulation-based performance analysis. The results demonstrate the system's viability for powering highway infrastructure such as street lighting and electric vehicle (EV) charging stations, thereby reducing reliance on the conventional grid and promoting sustainable transportation. Index Terms—Hybrid Power System, Vertical Axis Wind Turbine (VAWT), Solar Photovoltaic (PV), Maximum Power Point Tracking (MPPT), Renewable Energy, Highway Power Generation.

Design, Comparative Analysis, Simulation, and Hardware Test of a Five-Level Hybrid T-Type/H-Bridge Multilevel Inverter

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Abstract: To achieve more efficient and compact power conversion, hybrid multilevel inverter designs have attracted considerable interest. This study presents the design, comparative evaluation, simulation, and hardware validation of a five-level hybrid multilevel inverter that utilizes a hybrid structure integrating Ttype and H-bridge modules. By combining the advantages of both configurations, this method produces a superior five-level AC output using a single DC input, requiring fewer power electronic components than conventional inverter designs. The main goal is to create an economical and effective inverter that exhibits minimal Total Harmonic Distortion (THD), making it suitable for renewable energy systems such as photovoltaic installations. This research encompasses an in-depth investigation into the hybrid topology's operational theories, a comparative assessment against known designs such as Cascaded H-Bridge (CHB) and Neutral-Point Clamped (NPC) inverters, the formulation of an efficient Pulse Width Modulation (PWM) control method, and an appraisal of its overall performance. Crucial parameters including harmonic content, switching power dissipation, component numbers, and voltage strain are thoroughly analyzed to confirm the benefits of the proposed configuration.

Intelligent Cybersecurity Architecture for Flying Taxis Using AI-Based Intrusion Detection and Command Validation

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Abstract: Urban air mobility introduces new cybersecurity challenges as autonomous flying taxis rely on continuous networked communication and sensor data. This paper presents a Next-Generation AI-Driven Cybersecurity Framework for Flying Taxis that integrates real-time intrusion detection, GPS spoofing prevention, and secure command validation within a unified edge– cloud architecture. The backend employs a Random Forest-based Intrusion Detection System and an autoencoder-based GPS spoofing detector trained on simulated telemetry and attack datasets. RSA-based encryption ensures integrity and authenticity of remote flight commands. Implemented using a Flask backend and a React dashboard, the system achieved 96.2 % accuracy for network intrusion detection and 94.6 % detection rate for GPS spoofing, with an average response latency of 42 ms per request while maintaining <10 % CPU utilization. The proposed framework enables resilient, low-latency protection for cyberphysical aerial systems and can be extended to other intelligent transportation and defense-grade platforms.

IoT-Integrated Smart Charging Scheduler and Battery Management System for EV Battery Life Enhancement

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Abstract: Electric vehicles (EVs) require effective energy management systems to maximize battery life, safety, and cost efficiency. However, typical battery management systems (BMS) largely provide basic monitoring of voltage, current, temperature, and state of charge (SOC) without advanced decision-making capabilities. This study proposes an integrated IoT-based BMS with a smart EV charging scheduling. The proposed system utilizes an ESP32 microcontroller for real-time monitoring and incorporates thermal management and intelligent charging control to reduce battery stress and overheating. In order to minimise power costs, the proposed system uses an ESP32 microcontroller for realtime monitoring, thermal management, and intelligent charging control. A Time-of-Use (ToU) based scheduling strategy moves charging to low-tariff periods. Data analytics, remote monitoring, and real-time visualisation are made possible by cloud-based IoT integration. For next-generation EV energy management, the suggested system provides an economical, safe, and effective method.

A Novel LSTM-Augmented Deep RL Controller for Energy-Efficient HVAC Airflow and Damper Management

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Abstract: HVAC systems are a major cause of high energy consumption in buildings, but the traditional controllers like PID and Fuzzy-PID often have difficulties with the non-linear thermal dynamics, slow responses, and poor regulation of dampers. This research suggests a hybrid intelligent HVAC controller that will merge Long Short-Term Memory (LSTM) prediction with a Deep Reinforcement Learning (DRL) agent, driven by a mechanical airflow and damper dynamics model. The LSTM unit will focus on the short-term temperature and airflow trends, while the DRL agent (built on PPO/SAC) will use these predictions to calculate the best damper adjustments that are balanced between comfort, energy use, and actuator motion. The final evaluation of the hybrid controller against PID, Fuzzy-PID, LSTM-only, and RL-only baselines was conducted with the "HVAC Sensor Data for Dynamic Fuzzy PID Control" dataset. The system proved good at controlling room temperature, with RMSE down to 0.72 °C. The PID and Fuzzy-PID had values of 1.84 °C and 1.31 °C respectively. The overshoot was below 3% and the controlled system showed the best response among all other systems. The daily energy consumption was reduced by 27%, more than RL-only (17%) and Fuzzy-PID (9%). In addition to that, the damper activity decreased to 54% and the level of comfort increased to 97%. These results are the confirmation that the combination of prediction and reinforcement learning has provided an effective and energy-saving solution for the smart HVAC systems of the future.

**Quantum-Inspired RISC-Based Neuromorphic SoC Architecture for Federated Edge
AI with Data Driven Low-Power Digital ASIC Optimization in 6G-Enabled Smart-City
Electric Mobility Digital Twin Ecosystems**

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Abstract: This paper presents a silicon-validated 7 nm RISC-V-based neuromorphic System-on-Chip (SoC) designed for energy-efficient federated edge artificial intelligence (AI) in 6G-enabled smart mobility and digital twin applications. The proposed architecture integrates a programmable RV64 core with an event-driven spiking neural network (SNN) accelerator and on-chip learning support. A key contribution is a quantum-inspired scheduling and mapping framework that optimizes neuron placement and communication patterns, significantly reducing interconnect overhead and improving execution efficiency. The prototype SoC comprises 16 neuromorphic processing elements supporting up to 16K neurons with distributed synaptic memory. Post-silicon measurements demonstrate an average power consumption of 50 mW during inference, achieving 0.48 μ J energy per inference and sub-1.2 ms latency. Compared to state-of-the-art edge AI accelerators, the proposed system delivers up to 12 $\times$ improvement in energy efficiency under equivalent workload conditions while maintaining competitive accuracy (92.7% on benchmark SNN tasks). The architecture further supports on-chip federated learning with gradient compression, reducing communication overhead by 89% and enabling scalable edge deployment. A hardware-in-the-loop digital twin framework is implemented for electric mobility scenarios, demonstrating real-time adaptive optimization and improved system-level efficiency. The results establish the viability of combining neuromorphic computing, federated learning, and quantum-inspired optimization in a unified silicon platform, offering a scalable and energy-efficient solution for next-generation edge intelligence systems.



Track 4

Sensors, Actuators & Edge Intelligence

Aquavac: AI-Driven Autonomous Waste Collection for Enhanced Water Sustainability

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Abstract: The AquaVac is an autonomous waste collection robot presented for overcoming water pollution by identifying, classifying, and collecting waste efficiently from aquatic environments. The system combines computer vision, deep learning, and robotic automation to make real-time waste detection and segregation. The major elements of AquaVac are the YOLO (You Only Look Once) object detection algorithm, which finds and categorizes floating waste into biodegradable and non-biodegradable categories. This classification is provided into an automated waste segregation module managed by an Arduino Uno, servo motors, and relays and provides better waste sorting. There are two modes are in AquaVac: autonomous and remote-controlled. In autonomous mode, it exploits GPS, LiDAR, and cameras for navigating water bodies, avoiding obstacles, and optimizing waste collection paths. In remote-controlled mode, a human operator can intervene in complex environments like narrow water channels and heavily polluted areas. The robot uses conveyor belts, suction mechanisms, or robotic arms to collect waste and store it in a designated compartment. Moreover, sensor-based detection like gas and infrared sensors, improves the ability of the system for classifying waste with greater accuracy.

A Hybrid Security-Driven Optimization Architecture for Scalable Agricultural Wireless Sensor Networks

S. Swapna Kumar, K. Satyanarayan Reddy, Anju Vincent, Gayathri P A, Niranjan T, Madhav Manoj, Vyshnav P

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Abstract: Agricultural automation increasingly relies on Wireless Sensor Networks (WSN) and Internet of Things (IoT) systems to improve productivity, sustainability and environmental monitoring. However, large-scale deployments face challenges related to node energy depletion, data integrity, intrusion detection and secure decision-making. This paper presents a hybrid WSN IoT architecture that integrates K- Means clustering, PSO-based Cluster-Head optimization, IDS/IPS mechanisms and blockchain assisted data protection. The proposed approach aims to improve performance across energy consumption, detection accuracy, throughput, and privacy breach probability. Simulation and analytical evaluations across varied node densities (500–2000 nodes) confirm improved communication efficiency (up to 25% throughput gain), stable energy distribution (15% balanced load) and enhanced tamper resistance (privacy breach < 0.5%). Results demonstrate that the hybrid framework maintains scalable, secure operation suitable for modern digital agriculture.

EDUGUARD

Bestwin Paul, Bresto K Benny, Christo Sojan, Christopher Biju, Jismol Jose, Anusree K
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Abstract: Class bunking and student disengagement are common challenges in educational institutions, often leading to poor academic performance and reduced classroom effectiveness. EduGuard is an AI-powered system designed to address these issues through automated attendance tracking using real-time facial recognition. By detecting and recording student presence as they enter the classroom, it eliminates manual roll calls and prevents proxy attendance. In addition to attendance, EduGuard features a wander detection system that monitors students who leave classrooms without permission during lecture hours. If a student is found roaming outside the designated learning areas, instant alerts are sent to the concerned teacher and the Head of Department (HOD) for immediate action. This helps institutions respond quickly to unauthorized absences. Powered by technologies like Convolutional Neural Networks (CNNs), OpenCV, and deep learning, EduGuard ensures high-accuracy face detection and recognition. By combining AI surveillance with centralized attendance management, it promotes discipline, increases accountability, and supports a more engaged and effective learning environment

AI-Based Urban Noise Source Classification for Sustainable Smart Cities: A Deployment-Oriented Evaluation

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Abstract: Urban noise remains one of the most persistent environmental stressors in modern cities. While long-term exposure has been linked to sleep disturbance, cardiovascular risk, and reduced quality of life, most existing monitoring systems still rely on aggregate sound pressure levels that offer little insight into source-specific contributors. This paper presents a systematic evaluation of machine learning approaches for urban noise source classification, with explicit focus on the tradeoffs between accuracy and computational efficiency that matter for real world deployment. Using a dataset of 2,400 annotated recordings collected across diverse urban environments, we compare four model families including efficient neural architectures with data augmentation and three feature representations. We report results with confidence intervals across five random seeds to ensure statistical reliability. Our best-performing lightweight model, a Random Forest classifier trained on MFCC features, achieves $82.3\% \pm 1.1\%$ accuracy and processes audio segments in under 12 ms on embedded hardware. When properly tuned with augmentation, a MobileNetV3-Small model reaches $85.7\% \pm 0.9\%$ accuracy but requires $3\times$ the inference time on edge devices. We provide detailed analysis of failure modes, inference latency, memory footprint, and model calibration to support practitioners designing source-aware noise monitoring systems.

Multispectral UAV Sensing and Deep Learning Benchmarking for Weedy Rice Detection in Precision Agriculture

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Abstract: One of the main challenges in precision agriculture is recognizing weedy rice. It facilitates effective weed management and monitoring in paddy fields. Large agricultural regions can be efficiently monitored using high-resolution spectral data due to recent developments in UAV platforms equipped with multi spectral sensing systems. This research investigates deep learning-based semantic segmentation models to analyze multispectral UAV sensing data from the WeedyRice-RGBMS-DB dataset. The dataset includes spatially aligned RGB and multispectral images acquired by UAV-mounted sensors with pixel-level annotations. Several U-Net-based architectures are evaluated under identical experimental conditions to examine the effects of loss functions, input resolution, and attention mechanisms on segmentation performance. All models employ an early fusion strategy that combines seven aligned spectral channels. Standardized preprocessing and augmentation techniques are used to ensure a fair evaluation.

Experimental results consistently show that improved loss functions and architectural enhancements lead to better performance. The baseline U-Net achieves an IoU of 0.585, while the attention-enhanced U-Net performs best with an IoU of 0.681 and a Dice coefficient of 0.8059. These results highlight the effectiveness of multispectral sensing and deep learning for UAV based agricultural monitoring.

Hand Gesture Mouse With AI Support

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Abstract: A real-time, vision-based gesture-controlled mouse system is proposed for touchless human-computer interaction using a standard webcam. The proposed approach utilizes a landmark-based hand tracking framework to detect and track finger movements in live video streams. A set of geometric features, including joint angles and inter-landmark distances, is extracted from detected hand landmarks to determine finger posture and spatial relationships. Based on these features, gestures are classified using deterministic rule-based logic without relying on machine learning or training datasets. Recognized gestures are mapped to common mouse operations such as cursor movement, left click, right click, double click, scrolling, drag-and-drop, and screenshot capture. The system is implemented using computer vision libraries including OpenCV and a real-time hand landmark detection framework, and operates efficiently on consumer-grade hardware. Experimental evaluation demonstrates that the proposed system achieves low-latency and reliable gesture recognition under normal lighting conditions. Due to its simplicity, interpretability, and low computational overhead, the system is suitable for touchless interaction applications, assistive technologies, and academic human-computer interaction prototypes.

Real-Time Fraud Detection Using Sensor Data and Machine Learning for Secure Transactions

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Abstract: Fuel fraud at gas stations is a real problem that keeps happening. There is no way to prove it, but customers often pay for more gas than they get. This paper suggests a monitoring system that uses a camera, a flow sensor, and a proximity sensor that are all attached to the vehicle to record what happens when the vehicle is getting fuel. The sensors check how much fuel is going into the tank and if the nozzle was properly placed the whole time. The system sends an alert right away if the amount given out doesn't match what was paid for. A shared database keeps track of stations that are flagged multiple times, so other customers know to be careful. The system is cheap, easy to set up, and responds instantly.

IMAGEIN AI

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Abstract: In the current digital world, there is a growing need for efficient image manipulation systems based on intelligent web applications for the better functioning of creative industries and online platforms. Generally, image manipulation tools are associated with high costs of manual work, technical knowledge, and high-end hardware facilities, which makes the process labor-intensive and less accessible to common users. The introduction of the IMAGEIN AI system is based on an efficient and intelligent web application that offers AI-based image manipulation tools such as background removal, image restoration, generation of fill images, and recoloring of images based on cloud computing technology. The transformation tools of the IMAGEIN AI system based on Cloudinary APIs help users to upload images and perform desired operations to achieve high-quality output images in real-time. Additionally, the system offers efficient user authentication based on Clerk and a credit management system where users are provided with free credits and allowed to purchase additional credits via Stripe subscription models. This platform continuously manages user sessions, credit tracking, and transformation history, all while maintaining responsive user interactions through a cutting-edge frontend implemented using Next.js, Tailwind CSS, and other technologies. This way, IMAGEIN AI effectively carries out all the operations of AI-powered image processing and SaaS management.

Real-Time Attendance Tracking Using AI and YOLO

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Abstract: Attendance monitoring is an essential aspect of academic and corporate environments to ensure participation, discipline, and accountability. Traditional attendance systems, such as manual roll calls and biometric scanners, are often inefficient, error-prone, and require physical interaction. This paper presents a comparative review of recent studies on AI-based real-time attendance tracking systems using YOLO (You Only Look Once) and other deep learning models for automated face detection and recognition. Various research works are analyzed based on their methodologies, architectures, datasets, and performance metrics. The paper highlights the evolution from traditional to AI-driven attendance management systems and identifies the gaps addressed by the proposed model that integrates YOLO-based face detection, AI recognition, Django–Flutter architecture, and role-based dashboards for students, faculty, and administrators.

EVALMATE AI: An AI-Based Adaptive Assessment & Evaluation System

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Abstract: Today, modern education systems increasingly rely on digital platforms for conducting assessments, hence there is a growing need to automate and streamline academic evaluation with maximum efficiency, accuracy, and fairness. Conventionally, assessment and quiz creation processes are manual, repetitive, and time-consuming, increasing workload and reducing instructional productivity. The EVALMATE AI system is introduced as an intelligent web-based platform that integrates automated quiz generation, computer-based testing, and performance analysis using artificial intelligence techniques. It is EVALMATE AI's AI-driven question generation models and evaluation engines that assist in creating subject-wise quizzes and assessing student responses in real time. Along with this, the system continuously tracks student performance and provides analytical insights and feedback to both students and teachers, enabling timely academic intervention and improvement, thereby enhancing the overall effectiveness of the evaluation process. In this way, the EVALMATE AI system performs its AI-based assessment and analytics tasks. Through these procedures, EVALMATE AI reduces the dependency on manual evaluation, saves operational time and effort, and offers an efficient and adaptive assessment strategy. It serves as a practical solution for modern educational institutions and significantly improves the efficiency and quality of academic assessment systems. Through these procedures, EVALMATE AI minimizes manual effort, reduces operational overhead, and promotes a scalable and intelligent evaluation framework.

IoT-Based Radiation Leak Detection and Emergency Alert System for Medical and Industrial Labs

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Abstract: The growing application of radioactive materials in medical and industrial labs poses potential health threats to laboratory personnel and nearby communities, thus requiring immediate detection and response techniques. The existing radiation monitoring system is only responsive and cannot send real-time warnings and is therefore too slow in responding to critical situations. In this paper, an IoT-based radiation leakage detection and emergency notification system is proposed that combines wireless radiation sensors and cloud analytics to continuously monitor the environment and immediately identify threats. The proposed system uses a multi-sensor feature set that includes radiation intensity measurements ($\mu\text{Sv/hr}$), temperature, and humidity measurements, gas leakage measurements, as well as smoke sensor measurements. A two-layer machine learning architecture that combines an Autoencoder and an RF classifier is used to leverage radiation data and make decisions on normal laboratory activities and leakage situations, sending automatic alert notifications. Experimental analysis shows that it has an accuracy of 97.3% in detection and has further lowered the false positive rate to 2.1%, compared to conventional threshold-based systems. The proposed system has an average response time of 4.2 seconds from detection to alert delivery and further sends successful emergency alert notifications for 94.8% of cases within 15 seconds from threat confirmation. Long-term analysis reveals that the proposed system has an uptime of 99.2% over six months of continuous runtime, showing its applicability in radiation-sensitive environments. The system has been validated for six months using 12 IoT nodes, and it has shown high reliability, secure communication, and energy efficiency.

Cloud Based IoT Health Care System

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Abstract: Recent advances in computer and communication technologies have enabled remote healthcare services through continuous patient monitoring and diagnosis. However, existing systems suffer from issues such as unreliable data transmission, high latency, and complex data handling. This paper proposes a cloud-based IoT healthcare monitoring system that integrates smart device sensors with cloud platforms to improve real-time monitoring and diagnostic accuracy. The system collects telemetry data, including accelerometer, gyroscope, magnetometer, and GPS, directly from the patient and transmits it to the cloud using IoT Hub and Azure Cosmos DB. Real-time analysis using Azure Databricks enables timely detection of health changes and alerts. An Activity-Based Key–Value Mapping algorithm is used to classify patient activities accurately. Experimental results demonstrate reduced latency, improved data processing speed, and minimized data loss compared to existing systems. The proposed system supports applications such as activity monitoring, fall detection, sleep analysis, and remote patient tracking.

CareLink – An IoT-Based Smart Platform for Seamless Epilepsy Monitoring, Seizure Detection, and Patient Management

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Abstract: Recurrent seizures, a hallmark of epilepsy, make prompt diagnosis and treatment extremely difficult. Current systems frequently use data from a single sensor, which reduces their precision and capacity for prediction. By utilizing machine learning and multi-sensor data fusion, we create a real-time seizure detection and prediction system to overcome these constraints. Temperature, pulse oximetry, electromyography, and galvanic skin response (GSR) sensors are all integrated into the system and are linked to an ESP32 microprocessor. To identify seizures, threshold values were taken from the literature, and when thresholds were surpassed, warnings were sent to pre-specified contacts. In order to study seizure frequency, medical experts kept sensor data in a database. Potential future seizures were predicted by training a machine learning model using past data. For data visualization and real-time monitoring, the system was reachable through an online interface. Testing showed that the multi-sensor technique was effective, with 95% accuracy in seizure detection and 90% accuracy in seizure prediction. For better epilepsy control, this system offers a scalable and intuitive approach.

Nimbus: An AI Powered Smart Drone Tracking and Neutralization

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Abstract: The proliferation of Unmanned Aerial Vehicles (UAVs) creates many security issues in restricted and sensitive airspaces. This article describes an AI-based Drone Detection System called Nimbus, which utilizes Simulation-Based Predictive Interception Modelling. Nimbus is capable of detecting UAVs through a Multi-Sensor Detection Framework consisting of Optical Cameras and Radio Frequency (RF) Analysis, as well as a YOLOv5 Deep Learning Model trained on over 5,000 labelled drone images and publicly available UAV datasets. The YOLOv5 model was validated with 92.5% accuracy while providing real-time inference of less than 100ms/frame. In order to assist in planning a response, Nimbus also includes a Predictive Analytics Module based on Newtonian Motion Modelling, which considers the influence of environmental variables (e.g., wind) in estimating drone trajectories. To validate potential interception strategies (e.g. net capture or directed use of a countermeasure) prior to actual destruction and without jeopardizing the reliability and validity of the detection function, Nimbus utilizes a Simulation Environment based on Unity. The results of experimental evaluation from multiple simulated urban and restricted airspace scenarios found a significant increase in reliability of detection using Nimbus, along with an increase in the speed of response planning compared to traditional RF only detection systems. Nimbus provides a proactive framework for compliant airspace security through its combination of AI-based detection, trajectory prediction, and simulation validated neutralization strategies.

Smart Railway Safety System Using AI-Based Monitoring and Real-Time Incident Reporting

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Abstract: Passenger safety and security are the prime focus of rail networks, particularly in urban centres in India. The Railway Safety & Security Web Application represents a new way of using AI-based monitoring, real-time sensor alerts and user-friendly reporting by passengers to secure railways. The application employs a YOLO-based object detection algorithm that allows for webcam-based monitoring of train cabins so that incidents such as theft, medical emergencies, and suspicious behaviours can be identified. The proximity sensors are also designed to assist with preventing accidents due to improper positioning near train doors by sending alert messages to users standing closer to train doors. In addition, users can report safety breaches or medical emergencies through an easy-to-use Passenger Reporting Facility and Emergency Button to ensure that railway authorities can respond quickly with appropriate action. Frontend Development uses React.js and Tailwind CSS; Supabase serves as the database; Firebase provides real-time notifications. Through the services provided by Firebase, Supabase, and the development of the new application based upon a combination of AI-based monitoring, sensor-based safety features, and real-time reporting, there are significant improvements in overall rail security, with a more proactive and responsive approach to passenger safety.

Latency Optimized Application Placement for IoT Based Automated Anesthesia

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Abstract: Emergency medical IoT applications always demands an action with minimum latency. Fog computing reduces the lateness in actuation while comparing with the traditional cloud computing approach. In this work a latency optimised placement approach in fog computing is used for an automated anaesthesia application. A combined priority based and edgewards placement algorithm is utilised for latency reduction. This load balancing scheme tends to reduce latency, network usage, cost of execution of this application. Implementation was done in iFogSim. The implementation results show a reduction of 52% to 69% reduction in latency when compared with the traditional approaches. A fare reduction in usage of network and execution cost are also the notable outcomes of this placement approach.

Smart IoT-Based Terrace Gardening System

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Abstract: Urban terrace gardening offers a sustainable solution for improving food security and environmental quality; however, regular monitoring and maintenance remain challenging for working professionals. This paper presents an integrated smart terrace gardening system that reduces manual intervention for working professionals through automation and remote monitoring. The proposed system employs an ESP32 microcontroller interfaced with soil moisture sensors, an ultrasonic water-level sensor, and a flow sensor to continuously monitor irrigation conditions. Automated water delivery is achieved using a motor-driven submersible pump and solenoid-controlled nozzle valves, enabling demand-based irrigation. To support scalable deployment across terrace environments, LoRa communication is utilized between distributed sensor and actuator nodes. A camera module provides visual monitoring of plant condition. At the same time, Wi-Fi/GSM connectivity enables real-time data transmission to a cloud-based IoT dashboard and mobile application for user supervision and alerts. The proposed framework offers a low-cost, scalable, and sustainable solution for smart urban terrace gardening.

YOLO-Based Deep Learning with Attention Mechanisms for Robust Micro-Algae Identification

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Abstract: Micro-algae, vital for economic balance and economic sectors, present a challenge in detection due to their diverse size and condition. Although a CNN-based model has been used in numerous studies to address the increase of harmful algae, the YOLO model is thought to be more accurate in recognizing algae. This sophisticated deep learning technique is widely used to identify algae and categorize them. Several iterations of the convolution neural network (CNN) based on the You Only Look Once (YOLO) model were employed in this investigation. This paper summarizes the effectiveness of attention mechanisms for the classification of micro-algae.

IoT-Based Multi-Vital Signs Monitoring with AI-Driven Health Prediction Using ESP32-S3

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Abstract: Continuous health monitoring is important due to the increasing number of lifestyle-related and cardiovascular diseases and the lack of continuous clinical direction. This paper presents an IoT-based health monitoring system for real-time and low-cost measurement of vital health parameters outside the hospital environments. The Wearable biomedical sensors are used to measure the heart rate, blood oxygen saturation (SpO₂), heart rate variability (HRV), respiratory rate (RR), and body temperature. This sensor data is sent to ESP32-S3 using MAX30102 and LM35 sensors. A lightweight embedded neural network performs a real-time health risk assessment directly on the microcontroller without using any cloud services. This approach reduces the system delay, improves data privacy, and also increases reliability. The experimental results using real-time data shows better performance and effective continuous health monitoring.

IoT-Integrated Sleep Posture Monitoring for Health and Safety Using Velostat

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Abstract: This paper presents the design and development of an IoT-integrated sleep monitoring system that utilizes Velostat based pressure sensors to detect and analyse the user's sleeping posture. The system has four primary sleeping states, those are back, stomach, left side, and right side. Based on pressure variations from the belt, the sleeping position is detected. It is designed to help individuals with health conditions such as cardiovascular diseases or sleep conditions who must avoid specific sleeping postures. An ESP32 microcontroller is used to process the sensor data and transmit information through the Blynk platform via Wi-Fi. Users can also select unsafe postures, change alert timers, and receive notifications when unsafe postures are maintained for a certain period of time. A grace period of 20 seconds is used to prevent the false alerts caused by minor movements. This system offers a low-cost and user-friendly solution for reliable sleep position monitoring.

A Centralized Entry–Exit Duration-Based Facial Recognition Attendance Monitoring System for Universities

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Abstract: Precision in monitoring attendance remains a significant challenge in higher education institutions. Issues like proxy attendance, inefficiencies in manual tracking, and the lack of subject-wise monitoring systems contribute to this problem. This paper introduces a facial recognition-based centralized attendance monitoring system that uses a model for verifying entry and exit times. Unlike existing systems that count attendance based on a single detection, the proposed system records timestamps when students enter and exit the classroom. It also calculates the total time spent in class during the scheduled lecture. Attendance is granted only if the student meets the minimum required duration. The system integrates deep learning-based facial recognition, validation of subject-wise enrollment, and administrative control systems. The experimental results indicate improved reliability, resistance to fraud, and scalability for implementation at the university level.

Edge-Deployable Student Adaptability Prediction via Optimized Gaussian Naive Bayes with Resource-Aware Feature Engineering

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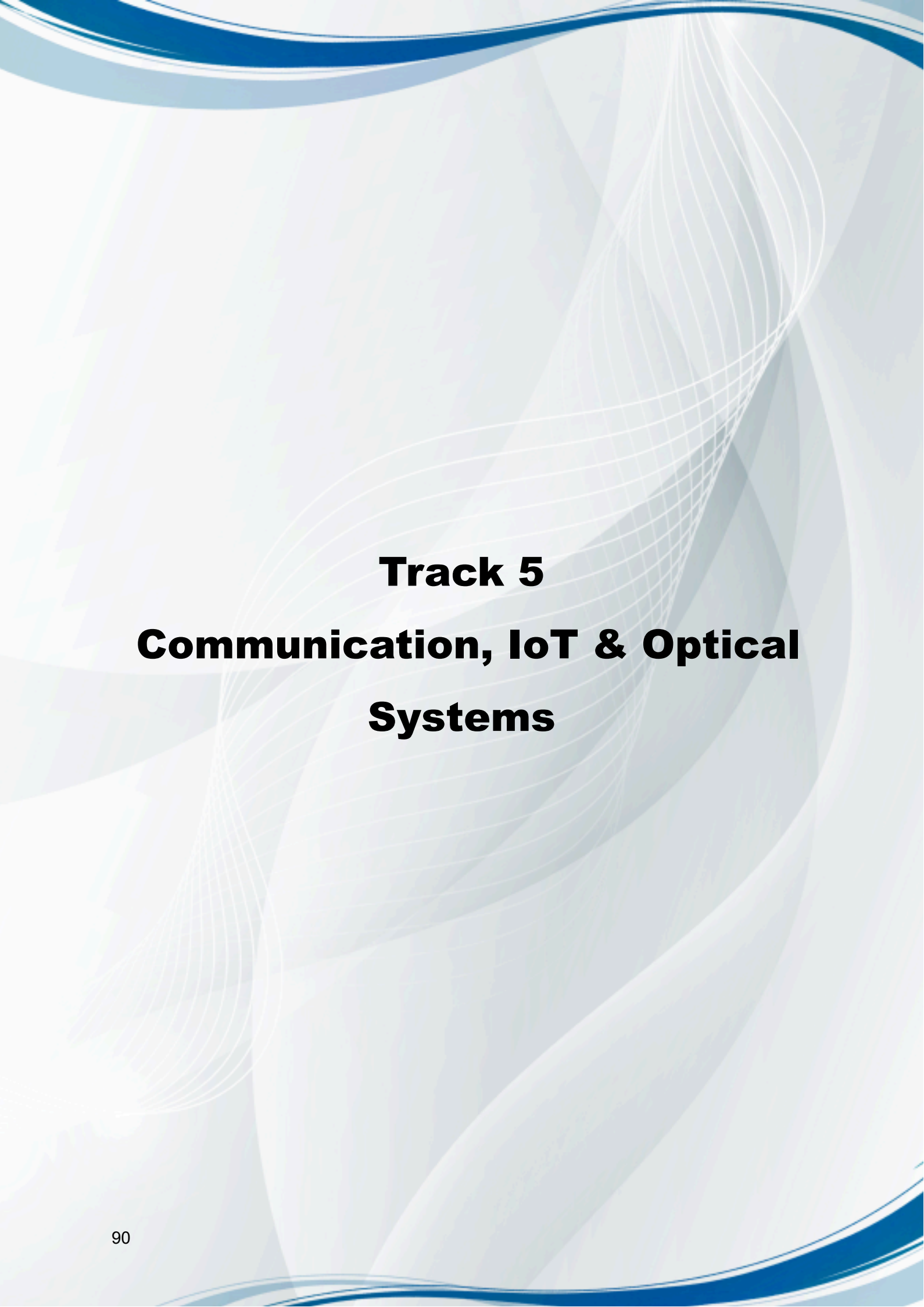
Abstract: When combined with a resource-aware preprocessing pipeline, Gaussian Naive Bayes achieves 86.31% accuracy while maintaining statistically comparable performance to a neural network baseline ($p = 0.443$) at $130\times$ lower training cost that enables real-time student adaptability prediction on edge devices without cloud connectivity. This paper presents an optimized Gaussian Naive Bayes (GNB) classifier trained on 1,205 student records with 13 behavioral and demographic features. Closed-form parameter estimation completes training in 0.88 ms, single-sample inference requires 0.091 ms, and the serialized model occupies less than 12 KB—fitting within ARM Cortex-A class processors without GPU acceleration. The proposed resource-aware preprocessing pipeline consisting of mode-based imputation, semantic-preserving ordinal encoding, standardization, and stratified sampling contributes significantly to model performance, which gives an 18-percentage-point improvement in accuracy compared with other unoptimized base- lines. Per-class AUC scores reach 0.932 (Low), 0.851 (Moderate), and 0.894 (High), macro-average 0.892. Permutation-based feature analysis identifies financial condition (importance = 0.122), load-shedding (0.041), and Self LMS access (0.039) as the three dominant predictors. Cross-institutional validation on the xAPI dataset ($n = 480$, Jordan/Kuwait) confirms pipeline generalizability, with GNB achieving statistically equivalent accuracy to the ANN baseline ($p = 0.880$, $\Delta = 1$ test sample) at $172\times$ lower training cost across both datasets.

Real-Time Multi-Parameter Cardiac Anomaly Detection System Using IoT and ESP32

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Abstract: This paper provides the hardware realization of a multi parameter cardiac monitoring system with real time based on the Internet of Things (IoT) technology. The system is based on a microcontroller, ESP32, and constantly measures five indicators of vital importance, including electrocardiogram (ECG), blood oxygen saturation (SpO2), blood pressure, blood glucose level, and physical activity. On-device edge processing categorizes the health of patients into risks. When abnormalities are detected the system sends SMS messages to a mobile device through a GSM module and generates instant alerts. This way, the data are displayed locally on an LCD and sent into a cloud dashboard to be monitored remotely. The viability of the decentralized healthcare monitoring system based on the prototype of the developed hardware is confirmed through the experimental evidence that the system can successfully identify the abnormalities and send alerts reliably, providing a high quality and low-cost solution.



Track 5

Communication, IoT & Optical Systems

Smart IoT-Based Framework for Wall Crack and Leak Detection Using Cloud-Integrated Lightweight CNN

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Abstract: Crack inspection is one of the critical measurement factors for the safety and durability of civil infrastructure. Conventional inspection techniques are costly, time-consuming, and often fail to detect small or concealed flaws. This paper describes a smart Internet of Things (IoT) system for wall crack and leak detection that combines cloud connectivity with a small, quick convolutional neural network (CNN). Early leak detection is made possible by the setup's use of an ESP32-CAM module for real-time wall image capture and a DHT22 sensor for humidity monitoring. The U-Net model performs pixel-level crack segmentation after data is wirelessly transferred to the Thing Speak cloud platform. The tuned ResNet50 network is used to classify the surfaces as cracked or uncracked. The U-Net demonstrated dependability in a variety of lighting and noise conditions with 75% accuracy, 91% recall, and 70% IoU. The modified ResNet50 model delivered accuracy and precision as 99.83% and 99%. The cloud dashboard generates instant alerts to facilitate timely maintenance decisions. The system is made to be inexpensive, energy-efficient, and suitable for both commercial and residential use. It offers a proactive and scalable maintenance plan that integrates deep learning and IoT.

An Optimized YOLOv8-Based Approach for Automatic Number Plate Recognition and Vehicle Sorting

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Abstract: Automated Number Plate Recognition (ANPR) is an essential component of contemporary smart transportation infrastructure, facilitating seamless vehicle tracking, access management, and traffic regulation. This paper presents a robust and real-time ANPR system that integrates the YOLOv8 object detection framework with EasyOCR for accurate number plate localization and character recognition. The proposed system captures vehicle images using a surveillance camera, where YOLOv8 is employed to detect and extract the number plate region with high precision. The extracted region is then passed to the EasyOCR engine, which performs text recognition to retrieve the alphanumeric characters. This two-stage pipeline ensures high detection accuracy, even under challenging conditions such as varying lighting, diverse plate formats, and partial occlusions. Experimental results demonstrate the system's effectiveness in achieving realtime performance with minimal computational overhead, making it suitable for deployment in smart city environments and automated toll or parking systems. By combining YOLOv8's powerful detection features with EasyOCR's lightweight, multilingual text recognition, the system achieves robust adaptability for diverse operational environments. Index Terms—Automatic Number Plate Recognition (ANPR), YOLOv8, EasyOCR, Object Detection, Optical Character Recognition, Smart Transportation, Real-Time Detection, Intelligent Traffic System, License Plate Detection, Deep Learning.

Energy-Aware Real-Time Rendering for Isomorphic Web Apps

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Abstract: A new, real-time, intelligent Server-Side Rendering (SSR) system for isomorphic web applications is proposed here. The objective of the research is to overcome the constraints of traditional SSR by employing Machine Learning to predict which web pages are likely to trend and therefore their active prerendering in order to significantly reduce the time of loading. The method utilizes a framework that dynamically adapts its rendering approach in real time as a function of real-time user activity, content type, user role, and device settings. It also includes a new cost-effective and energy-conscious decision layer that renders optimization activities by calculating power usage and estimate Return On Investment from user interaction. The performance shows major improvements in response time, server load, and noticeable energy savings compared to traditional Server-Side Rendering systems. Experimental results shown significant improvement in latency, server-resources usage, and energy efficiency, showing that the model does consistently deliver lower computational overhead and higher throughput than traditional SSR architectures.

Inset-Fed Microstrip Rectenna for Wi-Fi Energy Harvesting Using Voltage Doubler Rectifier

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Abstract: A compact 2.4 GHz rectenna integrating a voltage-doubler rectifier is developed to enhance Radio Frequency Energy Harvesting (RFEH) performance in low-power environments. The proposed rectenna utilizes a cost-effective FR-4 substrate with a low-profile microstrip patch antenna. The antenna employs an inset feed for standard 50Ω impedance matching, achieving a high maximum gain of approximately 4.16 dBi and a low return loss of -35.48 dB at the operating frequency. To ensure maximum power transfer, an LC impedance matching network was designed and optimized, achieving near-perfect matching. The rectifier stage utilizes a voltage doubler topology with Schottky diodes. The maximum RF-to-DC conversion efficiency achieved by the system is 61% at an input power of 10 dBm. Furthermore, the circuit exhibits stable operation, maintaining an efficiency above 20% for input powers exceeding -10 dBm, confirming high sensitivity for low-power harvesting applications. These results demonstrate a significant advance in compact, high-efficiency energy harvesting for powering low-power wireless devices.

SURAKSHA YATRA: An IoT and Edge-AI Based Cyber-Physical System for Offline and Online Tourist Safety

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Abstract: Tourism in ecologically diverse and remote regions frequently outpaces the development of reliable safety infrastructure. In topographies characterized by dense foliage or mountainous terrain, tourists routinely enter cellular "shadow zones," rendering conventional GSM-dependent emergency systems useless. To resolve this critical communication gap, this paper details Suraksha Yatra: a decentralized, cyber-physical safety architecture engineered as an EPICS initiative. The system directly addresses three core vulnerabilities in off-grid tourism: absolute network dependence, delayed anomaly detection, and the lack of verifiable incident data. The physical perception layer consists of an offline-first, ESP32- driven wearable that continuously processes inertial sensor data to classify physical anomalies, such as falls, directly at the edge. To bypass cellular dead zones, the wearable's communication stack dynamically fails over from a primary Bluetooth Low Energy (BLE) connection to a Long Range (LoRa) transmission protocol, guaranteeing distress signal delivery. On the backend, event logs are cryptographically hashed into Merkle Trees and anchored to a permissioned blockchain, establishing an immutable and privacy-preserving audit trail for emergency responders. This infrastructure is integrated with a Supabase and PostGIS backend to rapidly geofence hazardous areas, while a fine-tuned LLaMA-based Large Language Model handles multilingual emergency triage in real time. Extensive field trials conducted alongside the Andhra Pradesh Tourism Development Corporation (APTDC) validate the system's operational viability. Despite severe line-of-sight obstructions and intermittent connectivity, the hybrid transmission protocol sustained a 96.8% packet delivery rate. Furthermore, processing kinetic data directly on the microcontroller via a hybrid threshold-orientation algorithm yielded a 91.2% fall-detection accuracy. Ultimately, this architecture provides a highly scalable, fault-tolerant blueprint for modernizing tourist safety in infrastructure-poor environments.

**Incorporating privacy-preserving verification using Decentralized Identifiers (DIDs)
and selective disclosure mechanism**

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Abstract: The growing use of fake academic certificates and the ineffectiveness of the conventional verification means severely demonstrates the necessity of a safe, transparent, and large-scale solution. Traditional systems use centralized databases that are subject to tampering, loss of data and delays among administration. This article suggests a decentralized system of academic credential authentication based on blockchain technology and InterPlanetary File System (IPFS). Under the suggested system, educational institutions issue credentials by posting encrypted certificates to the Internet Protocol Framework, and store the matching content identifier (CID) on the blockchain using smart contracts. This guarantees permanency, integrity and effective record retrieval. By visiting the blockchain and getting related documents on IPFS, employers and academic institutions can check the credentials of students in real time and not require intermediaries. The prototype shows high security, transparency and faster verification than conventional methods and also consumes less storage overhead on-chain. The paper highlights how blockchain and IPFS combination can transform the process of educational records control and build trust in academic spaces.

Design and Structure of Autonomous Farming Robot

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Abstract: Most automated agricultural technologies are aimed at fruit harvesting, irrigation automation and crop monitoring. These mainly focus on maintenance phases of farming. The project aims to develop a data driven agricultural approach. Our project idea aims at developing a smart agriculture robot that plants the necessary seeds in designated places by leveraging technologies like autonomous SLAM based navigation systems utilizing the Robot Operating System(ROS) framework. The proposed system should be able to perform precise seed placement in designated fields with- out the need for manual guidance. Using ROS based packages the system should be able to perform real time mapping, localize itself using 2D/3D SLAM techniques using LIDAR/Cameras, Inertial Measurement Units(IMU). A Raspberry pi unit or an Intel NUC will be the main processing hardware to cater to the required needs. This would enable accurately spaced seed placement. Additionally ROS has tools like Rviz that helps pre testing before real time application. The hardware mainly consists of a robot chassis, movement system that consists of either general wheels or caterpillar tracks, a seed dispensing unit, DC motors, motor drivers, ultrasonic sensors and optional GPS. In future the system could be developed to have water control systems that work on real time data detected from the robot and also the robot could be developed using AI image detection techniques to monitor if the crop has any diseases. The system would also be able to support a user interface that provides real time data (sensor values and robot location) and a platform to control the robot from anywhere if necessary.

**Design and Execution of an EBG and SDR Driven
PIN-diode Reconfigurable Patch Antenna for Cognitive Radio Applications**

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Abstract: In order to make efficient use of electromagnetic radio spectrum, cognitive radio communication systems are utilized. The proper selection of the available frequencies requires an effective antenna. Cognitive radio (CR) systems are essential for dynamic spectrum management since the demand for effective spectrum utilisation has increased due to the quick expansion of wireless communication. Here, a reconfigurable microstrip patch antenna with PIN diodes and an Electromagnetic Band Gap (EBG) structure is designed and implemented. It is controlled by a Software Defined Radio (SDR) platform that uses GNU Radio. By changing the diode bias states, the antenna achieves frequency agility, allowing it to operate across several bands without requiring structural changes. Simulations in HFSS show that return loss is less than -15 dB, gain is greater than 5 dBi, and radiation efficiency is greater than 75%. Multi-band resonance and stable impedance matching are verified by a simulated prototype on FR4 substrate. Real-time spectrum sensing and dynamic frequency switching are made possible by interface with SDR hardware (USRP), confirming feasibility of adaptive CR communication.

Community Connect: A Django-Powered Platform to Improve Neighbourhood Communication and Resolve Local Issues

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Abstract: Local communities need to be well communicated with the ease of access to common resources in order to optimize their functionality. A good majority of the existing communication channels like informal messaging groups, physical noticeboards or generic social media platforms (e.g., Facebook) lead to fragmented information, lack of community involvement or no regulation of information. This paper describes a new web based system that will enhance local community interactions by providing a unified and organized approach to the management of communication and shared resources. The suggested system does incorporate role based access control that enables the administrators to verify and control user generated content, and develop a system that enables community members to post notices, attend events, borrow tools as well as trade books in an organization manner. The site can be scaled and is easy to use, using modern web technologies and a Django back-end and responsive front end design. The admissions by the administrative workflow processes that will approve the user generated content will increase the relevance and credibility of shared information. Results of functional testing indicate that this system will provide for increased access to community related information, improved coordination amongst members, and decreased reliance upon unstructured communication methods. Furthermore, the system has the flexibility to be used in any residential community (i.e., multi-family), hostel, as well as in institutional settings.

NETRA - An IoT Based Smart Glass System for Visually Impaired Individuals

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Abstract: People who are visually impaired in India and worldwide struggle to navigate safely and independently. Even simple things such as locating obstacles, traversing new places and keeping surrounding awareness can be difficult which can affect their freedom. A comprehensive solution is required to solve their problem. Netra is a Smart Glass System for Visually Impaired Individuals which aims to make their everyday life easier and accessible. It uses technologies such as Internet of things (IoT), Artificial Intelligence (AI), Computer Vision, Machine Learning (ML), Text-to-Speech (TTS), Global positioning System (GPS), Optical Character Recognition (OCR) and ultrasonic sensors that help them to improve their movement and independence while offering real time support. The system detects obstacles and alerts the users. Instead of just a beeping sound, it provides a verbal description of it. Netra helps users find direction and tracking locations. It provides the shortest route to their destination. It includes Weather Forecasting feature that updates the user about the bad weather. The system also supports text-to speech conversion,that helps the user to read and understand written materials.The system also aims to provide detailed descriptions about the surroundings and recognized people. This system integrates earphones to prevent others from hearing. In addition, an accompanying mobile app helps the carer to monitor and track the users. The carers also receive alerts in case of fall, emergency or if the user needs help. With the help of Netra, we can aim to provide safety, independence and quality of life to the visually impaired individuals.

A LoRa-Based Vehicle Anti-Theft System with Real-Time License Plate Verification

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Abstract: We present a robust, IoT-based vehicle anti-theft system that integrates a vehicle-mounted transmitter with a stationary computer vision receiver to validate a vehicle's identity. The in-vehicle unit, controlled by an ESP32 microcontroller, uses a LoRa module to broadcast the vehicle's true license plate number, a GPS module for location tracking, and a GSM module for SMS communication. The stationary unit employs a camera and an OpenCV-based Automatic License Plate Recognition (ALPR) algorithm to visually read the physical license plate. If the broadcasted plate number and the visually captured number do not match, the system triggers an alert, sending an SMS with the vehicle's location to the owner and enabling remote engine immobilization via a relay module. This digital handshake provides a proactive security layer against license plate spoofing and vehicle theft. The proposed system demonstrates high accuracy in verification and offers a significant improvement over traditional reactive security measures.

**Robust Data Watermarking and Verification Framework for Tabular Datasets
Using Teaching–Learning-Based Optimization (TLBO)**

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Abstract: With the exponential rise of open data exchange and machine learning applications, there has been an increase in the demand for a reliable method to protect ownership and verify the integrity of data. Although traditional cryptographic techniques provide confidentiality, they do not provide any way of preventing someone from modifying or using the dataset without authorisation. This paper proposes a Robust Data Watermarking and Verification Framework for Tabular datasets based on the Teaching-Learning-Based Optimization (TLBO) algorithm. The framework embeds a binary watermark into statistically stable attributes and optimally selects the location of the watermarks based on a fitness function that minimises distortion while maximising robustness using TLBO. The verification algorithm that is based on tolerance provides validation of the extracted watermark against various perturbation conditions. The experimental results provide evidence of a very good balance between the imperceptibility, robustness, and the ability to detect tampering. Future development of this framework will focus on hybrid TLBO-based methodologies and incorporation of a blockchain to create a watermark registry.

SIW Based Ultra Wideband Antenna for IoT and 5G Communication Systems

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Abstract: The high proliferation of Internet of Things (IoT) devices creates a need to use small, compact, high-performance, and wideband antennas, which enable the reliable work of multiple bands. The current paper describes a Substrate Integrated Waveguide (SIW) wideband antenna that is tailored to be used with the IoT. The design combines the better characteristics of the traditional waveguides and the planar microstrip designs which have a low profile, compact size, and low loss but can be readily integrated with the modern RF circuits. A wide range of frequency bands that are typical of an IoT system can be supported by optimization of the SIW cavity dimensions and attention to the technique of feeding, which provides a wide band of impedance. Metallic arrays of via adopt the radiation behavior of waveguide sidewalls and lower leakage. Full-wave results indicate that the radiation pattern, gain, and impedance were stable and comparable over the operation spectrum. The findings affirm that the proposed SIW antenna can be used to sustain high efficiency wireless connectivity in miniature IoT devices, which is a promising solution to the next generation IoT communication network wherein performance, reliability and miniaturization are paramount.

Comparative Performance Analysis of ML Algorithms for Credit Card Fraud Detection

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Abstract: As digital payments continue to surge, credit card fraud becomes an increasingly large source of financial loss or steep pitfalls for banks and consumers alike! Detecting fraudulent transactions can often be very difficult due to continuous evolution of fraudulent transaction patterns, increasing transaction volume, and extreme class imbalance between legitimate and fraudulent transactions. The goal of this research is to conduct performance comparison of a variety of different machine learning algorithms in regards to credit card fraud detection using a real-world imbalanced dataset of 10,000 credit card transactions. Implementations and assessments of logistic regression, random forest, gradient boosting, and XGBoost models were completed, using accuracy, receiver operating characteristic (ROC) area under curve (AUC), and F1-Score to provide a fair assessment under imbalanced conditions. Experimental results indicate that although logistic regression performs adequately in overall accuracy, it does not fare well regarding fraud-class precision. Conversely, ensemble methods improve fraud detection performance significantly, with both gradient boosting and XGBoost returning near perfect performance. Amongst the tested algorithms, XGBoost provided the highest level of performance, achieving 99.97% accuracy, 0.9995 ROC-AUC, and F1-Score = 0.9888. Thus, the results provide evidence for the efficacy of boosting-based methods when considering reliable and robust credit card fraud detection.

SCR-Auth: Secure Call Receiver Authentication on Smartphones Using Outer Ear Echoes

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Abstract: Smartphones allow users to receive calls even when the device is locked, which may expose sensitive information if the wrong person answers the call. To improve this security gap, this paper presents SCR-Auth, a secure and user-friendly call receiver authentication system based on outer ear acoustic characteristics. A whisper of sound, just beyond hearing, slides out the phone's speaker. Bouncing back, it returns as a faint echo caught by the mic inside. Each rebound shaped subtly by the curves of an ear. Hidden details emerge when broken down with MFCC methods. Identity checks happen by reading those acoustic traces. The system does not require additional hardware or explicit user interaction. Tested results show SCR-Auth reaches 96.95% on average for balanced accuracy - clear sign it handles real phone environments effectively. Built fast, yet still delivers solid results when used in practice.

Privacy-Preserving User Authentication Using Zero

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Abstract : In current times, technology is omnipresent; safely remaining logged in without endangering private information is no longer easy. Outdated methods such as static passwords and remembered credentials? Those are hacked much easier in current times because of phishing attacks, shared online passwords and shared session hacking, and massive business information breaches. But encrypted passwords are magnets for hackers looking for vulnerabilities instead. Here's what you can do instead: use Zero-Knowledge Proofs to prove yourself to the server without showing off the password at all. The system is based on a math puzzle exchange that leverages the basics of numbers and combined with genius hash solutions. By demonstrating oneself during account registration and login processes, this system is based on a simple web interface developed with Streamlit – permission management handled flawlessly in the background as well. No extensive business information storage like massive SQL databases; instead, this uses SQLite key storage for compactness and ease of use. Testing this system halts both leaked login information and man-in-the-middle attacks and imposter accounts with hardly any effect on speed either. It seems Zero-Knowledge Proofs are much more than theory; they actually do what they promise and keep eavesdroppers at bay for improved security.

Cyber Attack Prediction Using AI

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Abstract: The development of cyber infrastructure has now expedited, and accordingly, there has been tremendous growth in cyber attacks, which pose a threat to data security and even the integrity of networks. The traditional security approach, which consists of firewalls and signature-based IDS, faces limitations in the prediction of both unknown and emerging attacks. To mitigate the aforementioned problem, this paper proposes AI-based cyber attack prediction techniques that use ML techniques to scrutinize the patterns of network traffic datasets. The proposed approach identifies potential characteristics of the collected network datasets and marks them as safe or detrimental. To ameliorate the results during prediction and reduce the false positive rate, various preprocessing and feature selection techniques can be used. The performance evaluation of this proposed approach is done through the application of a publicly accessible standard dataset, and various performance measures like Accuracy Rate, Precision Rate, and F1-measure have been used to evaluate the assessment of the proposed cyber attack prediction approach respectively. To explore whether there exists any optimal approach for the existing proposed approach, comparative evaluation can be done on different ML-based classifiers. From the experimental results, it can clearly be seen that this AI-based approach is highly effective and accurate for unknown and known attacks. The approach possesses immense potential for mitigating the demands of real-time cyber security and therefore provides advanced security coverage for networks for the early warning of potential cyber attacks.

HADAP: Offline Parallel Anti-Virus Pipeline On Critical Infrastructure

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Abstract: Critical Information Infrastructure (CII) demands the use of malware detection systems which are fully offline, secure and auditable. The conventional antivirus products that had a single engine are not able to identify all the threats particularly the zero-day and polymorphic malware. The current paper introduces HADAP (Hybrid Antivirus Detection and Analysis Pipeline), a parallel multi-antivirus in that Windows Defender, Trend Micro Maximum Security, and ESET Internet Security are combined in independent virtual machines. HADAP is able to automatically scan files in the Target Folder, isolate infected files in a special Infected Folder, create detailed reports, and offer a central dashboard on which engines could be analysed individually. The framework also comprises VM snapshot and rollback services of infected files and safe cross-network scan and still maintaining complete offline compliance. Experimental analysis of one thousand files proves that HADAP has a coverage of detection of approximately 94, scan time is lowered by about 35 percent of sequential scanning, and total auditability is achieved under sensitive environments. HADAP is scalable, modular and appropriate to defence grade air-gapped networks.

Closing the Affective Loop in Human–Robot Conversation with Multimodal Affect-Aware Dialogue System and Adaptive Speech Expressiveness

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Abstract: Keeping people engaged in conversation is a core challenge for social robots and voice agents, especially in open-domain settings where users talk freely about everyday life. Many systems rely on text alone or react to a small set of emotion labels with fixed rules, which often misses subtle cues and can make responses feel repetitive or “robotic.” In this work, we propose an affect-aware dialog system that explicitly models user enjoyment and uses it to guide how the robot responds over time. We build our approach using the HRI CUES dataset, which contains conversations between older adults and a Furhat-based robot assistant, along with aligned transcripts, audio, face video, and human enjoyment ratings on a 1–5 scale. Our system encodes short context windows from three modalities—text, speech, and facial signals—into a shared dialogue-state representation. A supervised enjoyment regressor predicts the user’s current enjoyment, and this model is then frozen and reused as a reward signal for reinforcement learning. The learned policy selects both the response intent (e.g., question, answer, empathic reflection) and an emotional speaking style for text-to-speech (e.g., calm or cheerful), treating vocal expressiveness as part of the agent’s action space rather than a fixed setting. Offline evaluations with modality ablations show that multimodal perception improves performance compared to text-only input, with the full text+audio+visual configuration achieving the strongest results (93.10% accuracy, F1 = 0.91). We also find that allowing the policy to vary emotional TTS style across turns increases perceived expressiveness and supports faster recovery after low- enjoyment moments. These findings suggest that closing the loop around predicted enjoyment—using multimodal signals and speech style—can produce more engaging conversational behavior than simpler baselines, and can generalize to other engagement-focused agents beyond social robotics.

FPGA-based lip movement detection for improved hearing aid performance

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Abstract: For individuals with hearing impairments, relying solely on auditory signals often leads to reduced speech comprehension, especially in noisy environments. Lip movement detection provides critical visual cues that complement sound, enabling better understanding of spoken language. Recognizing this, the project proposes a novel approach to enhance traditional hearing aids by incorporating lip movement detection using Field Programmable Gate Array (FPGA) technology. The primary objective is to improve speech recognition for hearing-impaired individuals by supplementing auditory signals with visual cues derived from lip movements during speech. The methodology involves capturing images of lips pronouncing different alphabets and process them using proposed FPGA architecture. FPGA processes each pixel in a streaming fashion, performing: Gaussian filtering, Sobel gradient computation, Non-maximum suppression, Double thresholding, Adaptive thresholding, Feature extraction and Letter decoding. Further implemented in FPGA. The FPGA is programmed to analyse these binary inputs to identify and classify lip shapes corresponding to specific phonemes. By integrating visual and auditory information, the proposed system aims to enhance the accuracy of speech interpretation, particularly in noisy environments or for users with profound hearing loss. The expected outcome is a real-time, low-latency system capable of detecting lip movements with high precision. This research contributes towards the development of intelligent, multi-modal hearing aid technologies that promote better speech comprehension and an improved quality of life for individuals with hearing impairments.

Intelligent AXI4 Interconnect: A Reinforcement Learning Approach to Adaptive QoS Arbitration in Heterogeneous SoCs

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Abstract: We present Intelligent AXI4 Interconnect, a hardware RTL solution combining a SystemVerilog AXI4 crossbar with a lightweight reinforcement-learning (RL)– based arbiter for QoS. The design uses only combinational lookup-table (LUT)–based policy inference (no floating-point or matrix ops), enabling timing-safe integration in the critical data-path. The RL policy is trained offline and statically encoded, so only fast online inference occurs in silicon. We implement the crossbar and arbiter in SystemVerilog and perform full RTL-to-GDSII flow in a TSMC 16 nm FinFET process. Post-layout measurements show a maximum clock rate of 1.6 GHz, supporting a peak aggregate bandwidth of 102.4 GB/s. The intelligent arbiter eliminates head-of-line blocking and reduces the 99percentile latency by 55% versus a round-robin baseline. Under saturation, 94% of the bus is utilized, and overall performance improves by 30% compared to fixed-priority arbitration. The RL logic adds only <0.02 mm² of area (3.8% of the interconnect) and consumes 14 mW dynamic power, with no impact on critical-path timing. These results demonstrate that an adaptive, hardware- friendly RL arbiter can significantly boost QoS and fairness in heterogeneous AI SoCs while meeting strict area, timing, and power constraints.

System Architecture and Performance Analysis of a Spectral Feature Based Speech Enhancement Algorithm

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Abstract: This paper describes the system architecture and performance evaluation of a spectral feature-based speech enhancement algorithm that runs on the PYNQ-Z2 platform with the Zynq-7020 SoC. The proposed method uses short-time Fourier transform (STFT)-based time-frequency processing and reference-guided spectral feature extraction to make noise suppression more accurate. An adaptive noise estimation scheme controlled by voice activity detection (VAD) is used to keep noise statistics up to date, which allows the system to work well even when the acoustics are changing. A Wiener-like spectral masking technique is then utilized to reconstruct improved speech components while reducing musical noise artifacts. Unlike hardware-accelerated approaches, the complete algorithm is executed on the embedded ARM Cortex-A9 processor without programmable logic acceleration, thereby demonstrating pure software feasibility on resource-constrained systems. Experimental results show that the proposed system improves the input SNR from -4.18 dB to 2.66 dB. Perceptual quality and intelligibility metrics also improve significantly, with PESQ increasing from 1.62 to 2.31 and STOI from 0.71 to 0.82 . The average frame processing time is 3.8 ms, confirming real-time capability. These results validate the practicality and embedded efficiency of the proposed architecture for low-power speech enhancement applications.

IoT-enabled smart cloth peg: A novel approach to laundry management

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Abstract: The Internet of Things (IoT) is rapidly reshaping modern lifestyles by integrating smart systems into daily routines. With estimates suggesting over 75 billion IoT-connected devices by 2025, its applications are extending into diverse sectors, including the laundry industry. This paper introduces an innovative solution: the Smart Cloth Peg, a device designed to optimize the clothes-drying process through intelligent monitoring. Powered by solar energy for environmental sustainability, the device uses a moisture sensor to detect the dryness level of clothes. Once the clothes are fully dried, a notification is automatically sent to the user via a mobile application. This smart system not only enhances convenience and efficiency but also supports eco-friendly living through automation and renewable energy use. Communication between devices is achieved using TCP/IP socket- based data transmission over Wi-Fi.

Gesture-Controlled Hospital Bed Using Deep Learning and IoT for Patient-Centric Care

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Abstract: The advancements in computer vision and human-machine interaction has enabled the development of contactless assistive devices in the medical field. This paper presents the design and implementation of a gesture- controlled smart hospital bed system aimed to improve patient autonomy, hygiene, and convenience, especially for bedridden, elderly, or mobility-impaired patients. The traditional hospital beds are generally operated using physical remotes or manual controls, which is difficult to access, unhygienic, or uncomfortable for patients with limited movement. To address these limitations, the proposed system introduces a contactless, vision-based control mechanism with Raspberry Pi 4 for real-time hand gesture recognition using MediaPipe and a lightweight neural network, with recognized gestures transmitted via MQTT to an ESP32 microcontroller for motor actuation. Experimental results show a gesture classification accuracy of 99%, low end- to-end latency, and a cost-effective, contactless solution suitable for real-world hospital environments.

Power Efficient Implementation of Digital Receivers using Polyphase CIC Filter

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Abstract: In this work, a digital receiver has been designed and implemented to demodulate digital signals and extract the binary information they carried. For efficient down-sampling and decimation, a Polyphase Cascaded Integrator Comb (CIC) filter is employed. For eliminating the passband droop in CIC filters, the output of the CIC filter passed into an Finite Impulse Response (FIR) filter. Here, instead of using conventional multipliers, the method of Coordinate Rotation Digital Computer (CORDIC) is applied to realize the FIR filter coefficients, thereby dispensing with the requirement for hardware multipliers, reducing both power consumption and area. The digital receiver and the polyphase CIC filter worked together to ensure efficient signal processing. First, the incoming analog signal converted into a digital form using an Analog To Digital Converter (ADC). Once digitized, signal passes through the polyphase CIC filter, which performs decimation reducing the sampling rate while preserving all essential signal information. The polyphase CIC filter had effectively handled multi-rate signal operations by dividing the filtering process into several phases, thereby minimizing computational complexity and increasing overall power efficiency. Therefore, the proposed CORDIC-based method significantly reduces power consumption in the FIR filter architecture, achieving an estimated power of approximately 0.0820. Resource utilization also compared to highlight the area efficiency of proposed work.

Comparative Analysis of Energy-Efficient and Fault-Tolerant Clustering and Routing in Wireless Sensor Networks

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Abstract: Wireless Sensor Networks (WSNs) are widely used in data collection and monitoring applications, but node failures and limited energy resources severely limit their performance. A comparative performance analysis of fault-tolerant and energy-efficient clustering and routing techniques for WSNs is presented in this paper. An MGWO-based baseline method, a K-means clustering approach combined with Modified Grey Wolf Optimization (MGWO), an MGWO-based routing approach improved with Q-learning, and a hybrid approach combining K-means, MGWO, and Q-learning are all tested under the same simulation conditions. Performance metrics like network lifetime, packet delivery ratio, residual energy, and energy consumption are used in the comparison. The results of the simulation show that adding learning-based routing and clustering refinement enhances network lifetime, fault tolerance, and energy efficiency. For dependable and energy-conscious WSN operation, the study emphasizes the advantages of combining optimization and reinforcement learning techniques.

Remote Infant Care Using an IoT-Integrated Neonatal Incubator

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Abstract: Proper medical care during childbirth is essential, as delayed intervention and lack of neonatal facilities often lead to serious complications. In rural India, many newborns are delivered at home without access to trained medical personnel, which creates placing infants at high risk, especially those born prematurely or with low birth weight. To address this issue, a smart neonatal transport incubator has been designed to provide a stable and safe environment during transfer to hospitals. The system maintains the required temperature using an automatic control mechanism and continuously monitors vital parameters such as infant movement, body temperature, heart rate, and pressure. A LoRaWAN-based communication module allows real-time remote monitoring through a mobile application once the infant is safely placed inside in the incubator. Experimental results confirm reliable thermal regulation, with the incubator maintaining the desired temperature range and displaying a consistent value of 37.0°C. All sensor data is securely stored in a cloud database, supporting remote access, long-term analysis, and improved clinical decision-making. The proposed design can also function as a medical facility that provides remote services to hospitals, which is particularly useful in locations that are hard to reach.

Secure Low Latency Anesthesia Automation Using Fog Computing

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Abstract: Accurate and continuous anesthesia delivery is essential during long-duration surgical procedures to maintain unconsciousness, analgesia, and physiological stability, as improper dosing can result in intraoperative awareness or severe medical complications. Conventional manual administration and cloud-based healthcare systems are constrained by human dependency, high latency, connectivity interruptions, and security concerns, limiting their suitability for real-time, safety-critical medical control. Fog computing supports real-time processing and local decision-making by bringing computational resources closer to patients and medical equipment, making it an effective approach for automated anesthesia control. The effectiveness of such systems depends heavily on secure and efficient fog node placement to ensure timely dosage adjustments. Cloud and fog environments are analyzed using iFogSim to demonstrate the advantages of fog deployment, while Python-based simulations evaluate multiple placement strategies, including Genetic Algorithm (GA)-based, Reinforcement Learning-based (Q-learning), EdgeWard, Random, and Worst-Fit approaches using latency, network usage, and operational cost as performance metrics. Simulation results show that the GA-based placement algorithm provides the best overall performance, achieving approximately 80 percentage lower latency, 70 percentage reduction in network usage, and 70 percentage lower operational cost compared to non-optimized placement methods. The GA-based placement algorithm achieves a latency of 17 ms while reducing network usage by 5.00–74.67 percentage and operational cost by 7.14–71.74 percentage compared to existing placement strategies. These results validate the suitability of GA-based secured fog node placement for reliable, low-latency, and safe anesthesia automation in life-critical surgical environments.

Track 6
AI, Algorithms and
Sustainability in Urban
Systems

A Lightweight and Interpretable Machine Learning Approach for SMS and Email Spam Detection

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Abstract: The widespread use of short text communication platforms such as SMS and email has significantly increased the volume of unsolicited spam messages creating security risks and reducing user trust. Although several machine learning and deep learning approaches have been proposed for spam detection many of them are computationally expensive or operate as black-box models making them unsuitable for practical deployment where transparency and efficiency are important. This work presents a lightweight and interpretable machine learning framework for spam detection in both SMS and email messages. Separate processing pipelines are designed for SMS and email to account for their structural and linguistic differences while maintaining a unified framework. TF-IDF is used for text representation and efficient classifiers such as Logistic Regression, Random Forest and Multinomial Naive Bayes are combined using soft voting ensembles. For SMS messages additional handcrafted features are incorporated to capture informal writing patterns commonly observed in spam.

The proposed framework is evaluated using the UCI SMS Spam Collection and the Enron Email Corpus. Experimental results show that the model achieves an accuracy of 97.8% for SMS spam detection and 99.3% for email spam detection while remaining computationally efficient. Furthermore, the integration of LIME provides clear word-level explanations for individual predictions improving transparency and user trust. These characteristics make the proposed approach suitable for real-world applications such as telecommunications and financial services.

Focus Preservation Index: A Perceptually Aligned Metric for Multi-Focus Image Fusion Evaluation

Santhi Krishna M S, Harikrishnan D, Shiyas C R

CUSAT

Abstract: Multi-focus image fusion (MFIF) combines the focused regions of multiple partially focused images into a single all in focus image. In fact, the majority of current evaluation metrics including Structural Similarity Index Measure (SSIM), Entropy and QAB/F are general purpose measurement tools that indirectly imply focus quality and their performance is not always consistent with human perception in terms of sharpness retention. The Focus Preservation Index (FPI) is a novel objective metric for MFIF, aiming to assess the extent of focused information retained. FPI considers the uniformity of focus between the source and fused image by means of a gradient-energy map, local contrast discrimination and scale adaptive aggregation. Experiments on publicly accessible MFIF datasets showed that FPI highly correlates with perceptual focus retention, distinguishes fusion algorithms correctly, and performs more credible measurement than classical metrics. Findings validate FPI as a strong supplemental or stand-alone criterion for comparison of future MFIF research.

Phishing Website Identification Using Sparse Representation and Autoencoder

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Jawahar College of Engineering

Abstract: In recent years the internet has become an integral part of our day-to-day life. Nowadays all activities such as communication, shopping, and the transaction are done online. So large no of attackers or criminals should arise to steal confidential information of the user. The criminals view the internet as an opportunity to transfer their physical crime. One type of internet attack is a phishing attack. Most of the phishing attacks are happening through website or email. It threatens too many internet users particularly those who are not aware of the crimes on the internet. An original and phishing website has almost close and features resemblance. Both of them have minimal differentiation only. Most of the phishing detection method has less ability to differentiate this. Therefore, identifying a phishing website automatically is very difficult and accuracy is limited. This project aims to develop a sparse representation algorithm by jointly considering ‘task-specific’ relations between response variables and features and ‘self-representation’ relations among features in a sparse regression framework. By integrating these two different relations we formulate a new sparse linear regression model for class- discriminative feature selection. An autoencoder is used for selecting the unique features. The selected features are trained using an SVM to get a better classification. This method improves accuracy. And also compare the proposed method with PCA+SVM and PCA+RF. Results show superiority in the proposed method

An Intelligent Machine Learning–Based System for Automated Examination Hall Seating Arrangement in Universities

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Abstract: Preparing hall-wise seating plans for large-scale university examinations, particularly with numerous students, multiple subjects, and eligibility constraints, is traditionally labour-intensive, error-prone, and time-consuming. This study presents an automated seating allocation system that processes a dataset of appearing students, including register number, name, subject codes, eligibility status, and fee-payment status, and filters out ineligible or non-fee-paid candidates. Using Python libraries such as pandas and openpyxl, the system performs data pre-processing, groups students by subject, and applies a constraint-based heuristic algorithm to assign seats in examination halls with a capacity of 30. The algorithm ensures that no two students taking the same subject are seated adjacently, adheres to hall capacity limits, and balances student distribution across halls. The system generates hall-wise seating sheets in Excel or PDF format, listing seat number, register number, name, and subject code, along with a summary sheet for administrative oversight. Results indicate a significant reduction in manual effort, improved fairness in seat allocation, and compliance with institutional examination regulations, demonstrating the system's suitability for adoption by universities such as KTU to enhance the efficiency of their examination cells.

Home Companion Robo

Jeevan Jacob ,Joepaul Paulson,Levin Vincent,Roshan Peter Cherian,Ms. Athira K P,Ms.

Shobha Xavier

Jyothi Engineering College

Abstract: Mobility limitations turn even basic tasks—like picking up a water bottle—into a daily obstacle for many elderly and recovering patients. Caregivers end up playing “human fetch robot” all day, which isn’t exactly the best use of their time. To tackle this imbalance, we introduce Chitti, a low-cost indoor service robot built to handle simple item-retrieval tasks through voice commands. Using a Raspberry Pi 4, onboard vision, and a servo-based gripper, the robot identifies and fetches common household or healthcare items without relying on cloud processing. A smartphone or wearable interface allows hands-free control, making the system usable even for individuals with limited mobility. The design prioritizes affordability, modularity, and ease of deployment, making it suitable for small clinics, rehabilitation centers, and home environments. While Chitti isn’t trying to replace high-end humanoids, it fills the practical gap by automating repetitive caregiving tasks and restoring a measure of independence to patients. Future extensions include autonomous navigation, improved perception models, and integration with healthcare monitoring systems.

Hybrid YOLO-CNN Architectures for High-Performance Real-Time Waste Classification in Smart Waste-Management System

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Acharya College of Engineering Technology

Abstract: Automated garbage separation is a critical component of enhanced recycling rates, the protection of human health, and the realization of smart garbage management networks. In this work, we introduce a model of hybrid object detection and classification networks that combine the benefits of YOLO localization with CNN-based feature extractors in order to address both the quality and the processing speed of garbage classification. A Kaggle garbage dataset with YOLO labeling in six categories—biodegradable, cardboard, glass, metal, paper, and plastic was enhanced by cropping and transformations. Four architectures of increasing complexity were evaluated: YOLO-EfficientNetB4, YOLO-DenseNet201, YOLO-DenseNet201 with CBAM, and the optimized YOLO-DenseNet Fusion model. The baseline YOLO-EfficientNetB4 achieved 71.94% accuracy, CBAM integration improved it to 91.04%, and the final YOLO-DenseNet Fusion model further enhanced validation accuracy to 95.96% with an inference time of ~45 ms per image. Progressive improvements across models highlight the effectiveness of attention mechanisms and end-to-end feature fusion in capturing both local and global context. Extensive experiments, ablation studies, and confusion matrix analysis confirm robust class-wise performance and strong generalization. These results establish a reliable, low-latency AI framework for real-time waste classification suitable for smart city applications.

AI-Powered Fish Population Estimation and Coral Health Classification

Anagha Manikandan, Neethu T V, Ann Mariya Prince, Hanan K H, Hisana K H, Aswathy Wilson

Jyothi Engineering College

Abstract: Overfishing, coral bleaching, and habitat loss are threatening marine biodiversity. Underwater ecosystem monitoring by hand is time-consuming, error-prone, and labor intensive. This research suggests an AI-powered monitoring system that automatically classifies coral health problems and estimates fish populations from underwater photos using deep learning and computer vision techniques. The system is capable of processing real-time underwater data to identify fish, count them, monitor species diversity, and evaluate the condition of coral reefs by utilizing pre-trained Convolutional Neural Networks (CNNs) and object identification models such as YOLO. Researchers, marine conservationists, and government organizations can access real-time dashboards and data remotely thanks to the full solution, which is implemented as a web application using Flask or Gradio.

LO&FO: An AI-Powered Web Platform for Missing Person Identification and Recovery

Hrishikesh M S, James Shaju, Jeese John P S, Joel Dygy, Rehna Baby Joseph

Sahrdaya College of Engineering & Technology

Abstract: LO&FO is a web-based platform developed to tackle the rising number of missing person cases, including lost children, elderly individuals, and people with cognitive conditions such as Alzheimer's. It harnesses the power of artificial intelligence, with a facial recognition module driven by machine learning and image processing, to compare and match uploaded photographs with entries in its database, significantly speeding up and improving the accuracy of identifications. Users can submit detailed reports including photos, descriptions, and geolocation data of last known sightings, while the system provides real-time alerts on potential matches. Multilingual support ensures accessibility across diverse communities, and a community interaction module enables the public to actively contribute information, enhancing collective efforts. The platform also integrates secure authentication and encrypted data handling to protect user privacy. Designed for scalability and inclusivity, LO&FO supports collaboration between families, non-profits, and law enforcement, making it a powerful and socially responsible tool for reuniting missing individuals with their loved ones.

FairFarm: Attention-Driven MobileNet Architecture for Banana Leaf Disease Detection and Intelligent Selling Framework

Joice C Jomon, Mariya K B, Neha Paulson, Neha Johnson, Rehna Baby Joseph

Sahrdaya College of Engineering & Technology

Abstract: FairFarm is a smart farming framework that uses AI to help banana farmers spot leaf diseases early and sell their produce at fair prices. It combines deep learning-based image recognition with a smart selling platform to improve crop health monitoring and boost farmer income. In the system, farmers upload banana leaf images through the FairFarm application. The images are preprocessed and analyzed using a lightweight MobileNet model integrated with a Convolutional Block Attention Module (CBAM). The MobileNet backbone ensures efficient, real-time performance, while CBAM refines feature attention, improving the accuracy of disease classification. Upon detection, the system provides relevant disease information, preventive guidance, and connects the farmer to a smart selling module, enabling direct listing of healthy produce to potential buyers or investors. This lets them list healthy produce directly for buyers or investors. This shows that this mixed system holds great promise for scalable, sustainable, and farmer-friendly agricultural solutions.

Real-Time Tomato Leaf Disease Detection Using CNN and Web-Based Deployment

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Vidya Academy of Science and Technology

Abstract: This project presents an intelligent and accessible solution for rapid automated plant leaf disease diagnosis using deep learning. The system supports farmers, researchers, and agricultural workers by enabling fast and reliable disease detection without requiring specialized expertise. It addresses a key agricultural challenge where delayed diagnosis leads to disease spread, yield loss, and excessive pesticide use. The proposed system employs a Convolutional Neural Network (CNN) trained on a diverse dataset of 38 plant disease classes, with a focus on tomato leaf diseases. The model analyzes visual symptoms such as discoloration, texture deformation, and lesion patterns to achieve reliable classification, attaining a training accuracy of 88.8% and a validation accuracy of 82.6%, demonstrating effective generalization. For practical deployment, the model is integrated into a Streamlit-based web application that allows users to upload images for instant diagnosis. The system outputs the predicted disease, confidence score, and a ranked list of the top five predictions, along with disease-specific treatment recommendations. In addition, an AI-powered agricultural assistant built using the Gemini API provides interactive plant-care guidance and symptom-based support. By combining automated diagnosis with conversational assistance, the system reduces diagnostic time, cost, and reliance on expert intervention while promoting timely disease management. The results highlight the potential of deep learning and AI-driven interfaces in developing scalable and sustainable digital agriculture solutions.

Pitch Score: The Virtual Shark Tank Judge

Bincy Ann Marie Joseph, Nitika Achu Vinod, Pavithra Vijayakumar, Yazeen Meshal Sanuj,
Mahesh K M

Rajagiri School of Engineering and Technology

Abstract: The rapid increase in urban vehicle ownership has intensified parking congestion, resulting in increased traffic delays, fuel consumption, and environmental pollution. A significant portion of urban traffic is attributed to drivers searching for vacant parking spaces. Conventional parking management systems based on static sensors or manual monitoring suffer from high deployment costs, limited scalability, and maintenance challenges. In this paper, an Edge-AI-enabled Intelligent Parking Guidance System is proposed that integrates computer vision and Internet of Things (IoT) technologies to provide real-time parking occupancy detection and navigation assistance. The system employs deep learning-based vision models deployed at the edge to analyze camera feeds locally, thereby reducing latency and minimizing dependence on cloud infrastructure. Experimental evaluation indicates that the proposed approach achieves reliable detection accuracy with faster response times compared to cloud-centric solutions, making it suitable for large scale smart city applications.

Future Farmcast: A Survey on Agricultural Inventory and Market Forecasting

Abhinav K Manoj, Adithyakrishna J G, Al Ameen A K, Adhwai Shyjith, Dr. Neenu Johnson
Federal Institute of Science and Technology

Abstract: The community of farmers has its own issues, with selling products becoming increasingly larger and working together. The agricultural sector is observed to be under market pressure, with reduced surface yield and weakened trust. The survey paper provides a review of the systems in agriculture. Smart Agriculture Systems combine production forecasting and market analysis with community interaction systems. The survey paper considers papers found in Google Scholar, IEEE Xplore, Springer and ResearchGate. The study reviews a range of machine learning and deep learning techniques such as regression, decision tree, random forest, support vector machine, and neural network models. The proposed techniques are applied for estimating crop yield and for analyzing market trends based on data-driven methods. The survey also covers digital tools as farmer knowledge and communication, and co-decision making facilitator. Key shortcomings are the lack of real-time data integration, suboptimal prediction accuracy, poor consideration of climate variables and a usability problem. Based on the obtained results, the paper suggests that future research should focus on real-time data aggregation, better model accuracy and user-friendly systems adapted to requirements of farmers.

AI Powered Wildfire Prediction System

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Jyothi Engineering College Cheruthuruthy

Abstract: Wildfires constitute a significant threat to ecological stability, human safety, and resource infrastructure, particularly in climate-sensitive regions. This paper presents an AI-powered wildfire prediction system that is designed to predict fire surges and support proactive disaster management. The features like multi-source environmental data, including meteorological variables (temperature, humidity, wind), vegetation indices (NDVI), satellite-derived thermal anomalies, and historical fire records are integrated in this system. A novel and hybrid modeling architecture is being proposed here combining ensemble machine learning algorithms (XGBoost) with Long Short- Term Memory (LSTM) to capture both spatial and temporal dependencies in fire behavior. The preprocessing pipeline incorporates timeseries feature engineering normalization, and a rule- based annotation strategy that labels surge events based on future thermal anomaly spikes. The model gives the probabilities of daily surge and they are smoothed and categorized into severity levels for interpretability. These predictions are visualized through an interactive geospatial dashboard. This enables the real-time alerts, historical trend analysis, and district-level risk monitoring which can assist the authorities and take necessary measures. Model explainability is achieved using SHAP, and performance is evaluated using precision, recall, and F1-score metrics. The system is modular, scalable, and designed for future integration with additional data sources such as drone imagery or IoT sensors. The insights provided from this AI-driven wildfire prediction system, this platform empowers authorities to plan evacuations, allocate firefighting resources, and mitigate the environmental and human impact of wildfires.

High-Resolution Real-Time Environmental Monitoring System using ESP-NOW for Hyper-Local Air Quality Indexing

Abhiram K S, Joseph Mathew, Priya S, Samuel, Leander Figueiredo, Abhinav M B, Aju Mathew Thomas, Athul Martin

Rajagiri School of Engineering and Technology

Abstract: Air quality monitoring is critical for protecting public health. This paper presents an intelligent air quality monitoring and prediction system for a localized environment. The proposed system consists of a distributed IoT sensor network with machine learning algorithms for real-time monitoring and predictive analysis. This system is designed by employing ESP32 microcontrollers with a group of environmental sensors (PM2.5, PM10, VOCs, CO, NH₃, H₂, NO₂, temperature, humidity, and pressure). The device-to-device communication is done by the ESP-NOW protocol. Time-series data is processed using various models, including Random Forest, SVR, XGBoost, and LSTM networks, to predict future Air Quality Index (AQI). During experimental deployment the ensemble machine learning approach achieved a prediction accuracy of 97.2% ($R^2 = 0.972$) with an RMSE of 2.67, significantly outperforming baseline methods. Real-time validation showed 94.3% accuracy for 1-hour ahead AQI predictions, enabling proactive environmental management.

Crack Detection in RCC Beam Using Deep Learning and Image Processing Techniques

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Sahrdaya College of engineering and technology

Abstract: The strength and long-term usability of Reinforced Cement Concrete (RCC) elements often suffer from surface cracks. These cracks can be early signs of potential structural problems. Traditional methods for checking cracks by hand take a lot of time, are subjective, and are not efficient for large or hard to reach structures. To tackle these issues, this project introduces a Smart Crack Detection system for RCC beams that uses Artificial Intelligence (AI) and image processing. The system uses a deep learning object detection method based on the YOLOv11 (You Only Look Once) model to find cracks in high resolution images of RCC surfaces automatically. This system allows for quick, automated, and objective identification of cracked areas, greatly cutting down on human effort and inspection time. This AI-based method provides a non-destructive, cost effective, and scalable way to monitor structural health and detect surface defects in concrete structures early.

A Smart AI-Driven Framework for Automated Crime Reporting and Incident Management

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Malla Reddy Engineering College For Women

Abstract: Crime remains a serious social problem around the world, and traditional crime reporting systems are still traditionally fraught with inefficiency, delayed response time and lack of transparency. As a result, victims continue to be hampered by fear of retaliation, social consequences or lengthy bureaucratic processes that lead to underreporting of crime. The Automated Crime Reporting and Incident Management System (ACRIMS) proposed in this paper provides a digital solution to reporting crime and the incident management process using automation, safe digital storage, and anonymity. Keywords—Crime Reporting, Incident Management, Digital Policing, Smart Governance, Public Safety, Automation, Cybercrime, Kidnapping, Human Trafficking, Terrorism, Corruption, Drug Trafficking, Sexual Harassment, Homicide.

LEARNZO: An Explainable Hybrid Adaptive Learning Framework Using Contextual Bandits for Engagement-Driven Personalization in Neurodiverse Learners

Arya T, Athira K Jagadeesh,Devarsh C S,Caroline Rose Biju

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Abstract: Neurodivergent learners exhibit substantial variability in cognitive processing styles and engagement behaviors, which are often insufficiently supported by existing adaptive learning systems. Many current platforms rely on rigid rule-based personalization or opaque machine learning models, limiting both adaptability and pedagogical transparency. These limitations can lead to delayed adaptation, reduced interpretability of instructional decisions, and insufficient responsiveness to rapid fluctuations in learner engagement. This paper introduces LEARNZO, an explainable hybrid adaptive learning framework designed to address these challenges while maintaining transparency and ethical control. The framework initializes personalization through supervised inference of learning preferences using VARK questionnaire responses. Disability-aware pedagogical rules, defined by domain experts, constrain instructional actions to ensure accessibility and inclusivity. Instructional adaptation is formulated as a contextual bandit problem, enabling real-time strategy selection based on engagement feedback while explicitly modeling uncertainty. A LinUCB-based policy is employed to balance exploration and exploitation within the constrained action space, allowing dynamic adaptation without compromising explainability. Experimental evaluation using simulated learner sessions demonstrates a mean engagement improvement of 11.9%, outperforming non-contextual (6.1%) and static rule-based (4.8%) baselines. These results suggest that incorporating uncertainty-aware online decision-making enhances instructional responsiveness in engagement-sensitive learning contexts. The proposed framework offers a transparent and modular foundation for adaptive learning systems targeting neurodiverse populations.

House Price Prediction Using Support Vector Machine in Kerala: A Web-Based Approach

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Abstract: This paper presents a web-based House Price Prediction system designed for Kerala using a Support Vector Machine (SVM) model. It analyzes various property features such as location, area, number of bedrooms, proximity to city centers, and amenities to predict real estate prices. The system aims to improve transparency and data-driven decision-making in Kerala's real estate market. Built using Python, Django, and machine learning libraries, the system offers a simple user interface for real-time price prediction and planning. Experimental evaluation and simulation show that the system is both efficient and accurate in estimating residential property values.

LLM Integrated AI Assistant For Educational institutions

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Providence College of Engineering

Abstract: Through manual inquiry methods like phone calls, emails, and face-to-face visits, organizations face the problems of delayed responses, inconsistent information, limited staff members' access and greater workload. To counter these limitations modern inquiry automation systems were developed and empowered by AI chatbots that are built with Retrieval Augmented Generation (RAG) [8]. The organization uses this method to collect all its knowledge which gets transformed into a vector database for permanent storage. The system uses this system design to access needed data with high speed. The system validates answers through trustworthy data sources. The system achieves better accuracy through vector search which identifies contextually relevant data. The knowledge which has been collected after a user makes a request helps a small language model to produce highly accurate and steady responses. The systems operate continuously to handle multiple users while delivering dependable and current information. The system provides an AI-assisted inquiry automation solution which achieves both maximum scalability and complete future readiness.

Vision-Based Fire Detection Using Surveillance Cameras for Smart Safety Systems

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Kothapalli Sowmika, Mukka Aashritha

Malla Reddy Engineering College For Women

Abstract: Fires can kill people, destroy buildings, or harm nature. Spotting flames early helps limit destruction while speeding up emergency actions. This study builds a camera-based tool that spots fires automatically through image analysis plus smart algorithms. It checks real-time video, spotting things such as glow tones, shaky motion, or odd patterns. Besides that, pulling info from extra gadgets - like heat or smoke detectors - cuts down on fake alerts by using different signals instead. Tests show the new system spots fires fast, plus it's precise giving an effective alert option for homes, factories, or woods. The goal? Strengthen safety while also making emergency responses quicker using smarter methods that adapt on the fly.

VeriClassify: An AI-Driven Document Categorization within a Blockchain-Secured Git Workflow

Harry Thomas Binu, Joseph Hadlee, Jogin Simon, Diyon Saju, Dr. Neenu Johnson

Federal Institute of Science And Technology

Abstract: This paper presents a secure and scalable frame-work for intelligent repository monitoring and verification. The system integrates AI-driven classification with blockchain-based integrity anchoring to ensure tamper-evident record keeping. Instead of relying on complex custom cryptographic protocols, the current implementation employs secure token-based authentication, encrypted communication channels (TLS), and blockchain-backed immutability to guarantee data integrity and non-repudiation. Experimental results demonstrate the feasibility of the approach in real-world repository synchronization environments while maintaining security and performance efficiency.

ZeroPath APP: AI-Powered Waste Classification for Smart Cities Using EfficientNet

Dhiya Devassy, Milna Jose, Rahul Dhanaraj, Shins John Jaison, Dr.Sandeep C S

Jyothi Engineering College

Abstract: Urban waste management can be revolutionized with the help of the ZeroPath App, a new mobile solution. It enables direct online communication between consumers, recycling markets, and local government organizations. The application lets users categorize waste materials by using artificial intelligence via a convolutional neural network, particularly Efficient Net B3, for picture identification. They can capture photographs with their mobile devices. This sorting activates a smart system that maximizes collection routes, organizes pickups, and dispatches trash to neighborhood recycling facilities or non-governmental organizations. Additionally, the platform uses predictive analytics for demand forecasting, employs a gamified reward scheme to promote user participation, and provides real-time tracking to improve efficiency. ZeroPath supports communities, reduces environmental pressure, and helps in the development of sustainable smart cities by fostering responsible consumption, simplifying logistics, and promoting circular economy.

AI-Based Autonomous Coconut Tree Climbing and Harvesting Robot

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Kizhupadath Mohandas

Jyothi Engineering College

Abstract: Coconut harvesting is a labor-intensive and risky agricultural activity that depends heavily on skilled tree climbers, whose availability has declined in recent years. To address this challenge, this paper presents an AI-based autonomous coconut tree climbing and harvesting robot designed to improve safety, efficiency, and productivity. The proposed system integrates a motor-driven climbing mechanism with adaptive grippers and a computer vision module based on the YOLOv8 deep learning algorithm to detect mature coconuts in real time. A Raspberry Pi 5 is used as an edge computing platform to process visual data and control robotic operations. Once a mature coconut is detected, a servo-controlled cutting mechanism is activated to perform selective harvesting. Experimental results show that the trained model achieves a detection accuracy of approximately 85% under varying environmental conditions. The proposed system reduces human risk, minimizes labor dependency, and offers a smart and sustainable solution for modern agricultural practices.

Decentralizing Merit: A Framework for Fair Content Evaluation on Blockchain Platforms

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FISAT, Angamaly

Abstract: Blockchain-based Online Social Media (BOSM) platforms have emerged as decentralized alternatives to traditional Web2 social networks, addressing long-standing concerns regarding censorship, data ownership, and unequal distribution of economic value. Token-based incentive mechanisms are central to these platforms, enabling financial rewards for user contributions while promoting a creator-driven digital ecosystem. However, empirical observations indicate that these systems are vulnerable to manipulation, including bot-driven voting, collusive behavior, and reward concentration, which weakens the intended correlation between content quality and financial incentives. This paper evaluates the reward dynamics of BOSM platforms by analyzing the interplay among content quality, tokenomics, and user behavior. It further integrates insights from blockchain governance, privacy-preserving mechanisms, decentralized social network usability, and token-economy design to propose an improved quality-aware and manipulation-resistant reward framework. The findings demonstrate that existing systems fail to consistently reward high-quality content, emphasizing the need for more transparent, equitable, and machine-learning-integrated reward models.

Language Learning Paradigms: A Scholarly Overview

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Abstract: This is an overview from 42 papers in between the year 2002 to 2024 analyses a mock-up for an AI integrated language learning system to aid students academic and proficiency focused test preparation. The system is designed to be a web-based platform that includes NLP and speech processing, as well as adaptive learning techniques. The LLM environment of such portals is intended to offer grammar and spelling checks, interactive discussion-place training, listening based exercises, and reading-comprehension units according to standardized (language-test) models. The architecture will integrate a multi model NLP pipeline, speech-to-text processing, and AI-driven feedback to facilitate personalized learning. Since it is a preliminary concept tool, the main interest is in its design and working methodology and the consequences of an ultimately implementation. The platform is expected to improve language proficiency through real-time guidance, automated evaluation, and progress visualization.

Smart Autofocusing in In-Line Holography applying a Deep Regression Model

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SRM University

Abstract: Digital in-line holography allows for small, label-free 3D imaging. However, it is still hard to get fast and reliable autofocusing because of noise, twin-image artefacts, and the high computational load. This paper presents a data-driven autofocusing framework that combines deep regression learning with hologram enhancement based on compressive sensing. The High-Speed Iterative Shrinkage Thresholding algorithm is used to suppress twin images for the first time, which improves the accuracy of the reconstruction. Then, a stacked autoencoder is used to learn how to tell the difference between different defocused holograms, and sparse Bayesian regression is used to find the best propagation distance. Support vector machine-based recursive feature elimination reduces feature redundancy, which leads to a small and useful set of features. Tests on holographic blood cell images show that the suggested method can find the focus much faster than traditional sharpness metrics and also cuts down on inference time compared to current deep learning methods. The system has a high level of reconstruction accuracy and a low level of estimation variance, which shows that it can handle noise and defocus. These results show that the suggested method is good for fast and accurate autofocusing in digital in-line holographic imaging

COACH-X : Sequential Interview Performance Analyzer

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Abstract: Many candidates find it difficult to develop these abilities through regular practice, despite the fact that interviews require clarity, emotional control, and constant confidence. The majority of AI-based systems currently in use only offer one-time feedback and are unable to record how a user's communication style evolves over time. We offer a customized, multi-session interview coach that monitors a candidate's development over several simulated interviews in order to address this. The system analyzes how emotion, sentiment, and confidence indicators change over time using a BiLSTM sequential model, transformer-based sentiment analysis, and Speech Emotion Recognition (SER). Through the processing of DistilBERT-derived text embeddings, CNN-BiLSTM emotion representations, and MFCC features, the framework provides insightful, session-wise feedback that identifies strengths and enduring flaws. Compared to single-session evaluation, experimental results demonstrate strong performance in modeling user-specific trends and offering more trustworthy insights. All things considered, this work presents an adaptive, progression-aware interview coaching system that promotes better interview preparation and improves self-evaluation.

Predicting Stock Market Movements Using News Sentiment and Time-Series

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Sree,Kasichainula Anishwa

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Abstract: Because of the effects of both past trends and major external news events, accurately forecasting movements in the stock market proves difficult. This project combines more than one technique to enhance its predictive capabilities as one part examines past price trends like openings, highs, lows, close prices – in addition to monitoring volumes to notice changes in the attitudes of the market. At the same time, it derives sentiment indicators from financial headlines using language-processing software. This project does not confine its prediction capabilities to merely one technique as it directs those sentiment indicators together with number trends toward smart algorithms that infer predictions on probable price directions the market may take. This algorithm enhances the visualization of real market trends as it integrates numbers along with public sentiment on the desired trend or direction.

Seatmate: Exam venue details scheduling system for efficient seat allocation.

Manoj M, Anjana S, Swathi Krishnan P, Abhishek K L, Hijas K

Jawaharlal College Of Engineering And Technology

Abstract: Provides students information about the exam location, including room numbers, locations, and seat assignments. Any changes are communicated to the invigilators and students. It automatically assigns students using room management and scheduling software. It utilizes scheduling and room management software for the allocation of students to rooms based on capacity while drawing data from student information systems (SIS). Additionally, it guarantees prompt notifications, which lessens misunderstanding. Makes the exam process smooth, well-organized, and stress-free. Following allocations, the system creates unique exam information for every student, including assigned seat numbers, exam date, subject and room numbers. An integrated notification system that facilitates email, SMS, and app-based alerts is used to deliver these details. To ensure transparency and lower the possibility of misunderstandings or missed tests, students and invigilators are promptly informed of any changes, including changes to the venue or schedule.

CNN-Based Deepfake Image Detection Enhanced with Fourier Transform Features

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Abstract: Recent advances in deep learning have eased the development of very realistic deepfakes images with severe implications for digital security and media verification. Deepfakes can be used for creating misleading information, performing identity fraud, and causing malicious manipulation activities, thus ensuring the development of reliable verification tools. The use of advanced deepfakes for generating images with hardly any observable defects increases the difficulties associated with the development of verification tools through conventional methods. Traditional tools for deepfakes verification include those specifically designed or those based largely on convolutional neural network (CNN) designs with a strong focus on identifying pixel-level details in space. Despite the effectiveness of CNN-based designs for verification tool development, there are potential difficulties associated with the use of very realistic deepfakes images with largely ignored space-domain defects due to upsampling and image generation. To overcome the problems associated with poor verification tool development due to improved deepfakes images," The proposed CNN-based framework for deepfakes image verification incorporates features based on the use of the Fast Fourier Transform (FFT) for identifying space-domain defects ignored due to very realistic deepfakes images. As the combination of spatial and frequency domain features is incorporated in the same model, the approach proposed improves the robustness and accuracy of the detection. The approach is applied to publicly available deepfake image datasets on the basis of the performance criteria such as accuracy, precision, recall, and F1 measure.

Integration of Emotion-Aware Multilingual NLP and Persuasive Dialogue Modeling for Context-Aware AI Voice Agents

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Abstract: This project presents the development of a Multilingual AI Voice Agent designed for automated, context-aware lead follow-up using Natural Language Processing (NLP) and Emotion Recognition models. The system bridges communication gaps in sales and customer engagement by enabling two-way, human-like voice interactions in the customer's preferred regional language. It integrates Automatic Speech Recognition (ASR) for speech-to-text conversion, a fine-tuned IndicBERT model for dual intent and emotion detection, and a Dialogue Manager following the PESO (Pain–Emotion–Solution–Offer) communication model to generate empathetic, persuasive responses. Using Text-to-Speech (TTS) synthesis, the agent responds naturally, adapting tone and content to the user's mood and intent. A modular architecture combining ASR, NLP, TTS, and CRM integration ensures scalability and real-time adaptability. The system also features an Admin Dashboard for analyzing conversation logs, call histories, detected intents, and emotions. By supporting multiple Indian languages, the proposed solution promotes inclusive digital communication and aligns with UN Sustainable Development Goals (SDG 8, 9, 10, and 12)—fostering economic growth, innovation, reduced inequalities, and responsible automation. The project demonstrates a step toward emotion-aware, multilingual conversational AI for enterprise-scale customer engagement.

Agentic AI Systems for Flood Management: A Comprehensive Review of Prediction, Evacuation and Relief Coordination in Kerala

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Abstract: Flood disasters pose a persistent and evolving challenge to Kerala, a region distinguished by its dynamic monsoon climate, diverse geography, and vulnerable communities. This review explores recent advances at the intersection of artificial intelligence and flood management, focusing on the potential of deep learning models and agentic AI systems to reshape both prediction and disaster response. Through critical analysis of current literature, the review highlights how architectures such as transformers and GRU networks are advancing real-time flood forecasting, enabling more accurate early warnings that are invaluable for risk mitigation at the local level. In parallel, the rise of agentic AI—especially multi-agent systems offers a pathway to smarter, decentralized evacuation planning and coordinated rehabilitation strategies. By facilitating communication among autonomous agents and integrating multilingual support, these approaches hold promise for enhancing inclusivity and equity in disaster relief efforts. While existing systems tend to excel at pre-flood warnings, the fragmentation of post-disaster recovery in Kerala underscores the urgent need for intelligent, adaptable, and community-focused solutions. The concepts surveyed in this paper point toward a promising future where hybrid deep learning and agentic AI frameworks not only improve technical performance but also strengthen the capacity of local administrations and citizens to navigate flood events and their aftermath. This direction in research invites continued exploration, collaboration, and innovation, with the goal of building a more resilient Kerala, prepared to meet the challenges of flood disasters through the transformative power of intelligent systems.

VESA: A Secure Offline Voice-Controlled Smart Home Ecosystem Using Agentic AI

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Abstract: VESA presents a secure, fully offline voice-controlled smart home automation ecosystem that prioritizes privacy, reliability, and intelligent on-device operation without cloud dependence. The system integrates a lightweight speech processing pipeline comprising Whisper-Tiny for accurate speech-to-text transcription and a Kaldi-based ASR engine for continuous wake-word detection with low latency. Natural language understanding and decision-making are achieved using a locally hosted, quantized large language model deployed via an optimized inference server, enabling real-time reasoning on consumer-grade CPU hardware. A two-stage recognition framework ensures robust command interpretation, achieving high accuracy across multiple automation tasks. Context-aware control is facilitated through an agent-based reasoning and acting framework that supports multi-step decision execution and conditional queries. To enhance personalization and robustness, the architecture incorporates retrieval-augmented generation using local knowledge bases along with layered fallback mechanisms and comprehensive error handling. Seamless IoT device control is realized through custom serial communication protocols. Experimental evaluation demonstrates reliable performance, fast response times, and efficient resource utilization, requiring minimal memory. VESA validates the feasibility of deploying advanced conversational AI and intelligent home automation entirely offline while maintaining strong security, privacy, and operational dependability.



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